A Project Report on

INTEGRATING EFFICIENTNET-B0 AND VISION TRANSFORMERS FOR ENHANCED GLOBAL AND LOCAL DIAGNOSTIC REASONING

is submitted in partial fulfillment of the requirement for the award of the Degree of

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G. PULLAIAH COLLEGE OF ENGINEERING AND TECHNOLOGY

(Autonomous)

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Pioneering Innovative Education

DEPARTMENT OF DATA SCIENCE

G. PULLAIAH COLLEGE OF ENGINEERING AND TECHNOLOGY

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CERTIFICATE

This is to certify that the project report entitled **"Integrating efficientnet-b0 and vision transformers for enhanced global and local diagnostic reasoning"** being submitted by **Golla Mounika(22AT1A3211), Arshiya Tarannum Shaik (23AT5A3207), Jhansi Lakshmi Mandugula (22AT1A3228), Harshini Vidapanakallu(22AT1A3247)** in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Data Science of G.Pullaiah College of Engineering and Technology, Kurnool is a record of bonafide work carried out by them under my guidance and supervision.The results embodied in this project report have not been submitted to any other university or institute for the award of any Degree or Diploma.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

We hereby declare that this project work entitled **"Integrating efficientnet-b0 and vision transformers for enhanced global and local diagnostic reasoning"** in partial fulfillment of requirements for the award of degree of Computer Science and Engineering – Data Science is a bonafide work carried out by us during the academic year 2024-25.We further declare that this project is a result of our effort and has not been submitted for the award of any degree by us to any institute.

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ABSTRACT

Due to an increase worldwide in skin cancer cases, it is critical to develop new diagnostic tools that assist physicians in making accurate diagnoses during routine visits. Many of the currently utilized CNN models have a high level of effectiveness in image detection; however, most CNN models only focus on local elements of the images such as textures and edges of the images. In addition, many CNNs cannot take into consideration the larger piece of context needed such as: the overall symmetry of the structures and the relative locations of irregular border structures, which are two important characteristics in dermatological diagnosis. Consequently, this proposed project will offer a hybrid deep learning architecture that will provide an optimal structure for the classification of seven different types of skin lesions on the HAM10000 dataset.The new system uses a combination of EfficientNet-B0 and Vision Transformer (ViT) to handle Local and Global image features at the same time. A 1x1 convolutional bridge transforms Local features into Latent Variables that can be used within the Transformer Blocks function of the model to produce a full Spatial Context of Input Data. This Combination Method enables our model to have an understanding and see tiny Cellular Defects while also being aware of the overall size of the Lesion. The newly written architecture produces a Diagnostic Accuracy Level of 87.2% and provides superior features in distinguishing between Benign and Malignant Cases when compared to traditional Stand- Alone Models.

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1

INTRODUCTION

INTRODUCTION

1.1 Overview of Computer-Aided Diagnosis in Dermatology

There is a worldwide epidemic of skin cancer, With the number of people suffering from skin cancer being at its highest levels ever, it is essential that computer-assisted diagnostic (CAD) systems accurately detect skin cancer during its early stages. Unfortunately, traditional convolutional neural networks (CNNs) are typically constrained to local receptive fields that focus solely on textures, therefore unable to detect complete structural symmetry or holistic view of lesions as required by clinical diagnostic rules such as “the ABCD rule” which depend on global view vs. local filters. This research aims to address this issue through the integration of a hybrid model based on EfficientNet-B0 and Vision Transformers. Through this integration, the model can capture both minute patterns; and also structural integrity on a more global scale than traditional convolutional neural networks. Hence providing an overall better diagnostic investigation of skin lesions than would be possible using only CNNs.

The technological innovation is based on a 1×1 convolutional bridge that connects the CNN feature maps with the Transformer tokens. This allows for the retention of the spatial richness of the CNN feature maps while providing access to the global reasoning of the self-attention mechanism. The study uses the HAM10000 dataset with weighted sampling to address the class imbalance; thus, avoiding model bias towards the common benign class while maintaining the highest level of sensitivity for malignant classes. The evaluation of the model is performed using the Macro-averaged F1-score to provide a fair comparison across all categories. This methodology addresses reviewers’ comments by focusing on diagnostic fairness and replication of research.

The system provides an even higher level of ease in documentation via the request for standardized PDF reports that contain the results of AI-based assessments. Through Attention Mechanism integration with user-centric deployment, the

project provides an innovative and ground-breaking interface for contemporary dermatology. The development of the state-of-the- art interface will allow for access to cutting-edge AI technology by providers who are not specialists in dermatology, thereby enabling earlier screenings and better long-term health outcomes for patients with Dermatologic conditions.

The rapid advancement of Artificial Intelligence, particularly in the field of Deep Learning, has significantly transformed modern healthcare systems. Among various medical domains, dermatology has greatly benefited from automated image-based diagnostic tools due to the visual nature of skin diseases. Early detection of skin cancer remains a critical challenge worldwide, as delayed diagnosis can lead to severe complications and increased mortality rates. In this context, Computer-Aided Diagnosis (CAD) systems have emerged as powerful tools that assist clinicians in identifying skin lesions accurately and efficiently. These systems aim to reduce human error, improve diagnostic consistency, and provide faster preliminary analysis, especially in resource-limited environments.

Deep Learning plays a crucial role in enabling such intelligent diagnostic systems. Unlike traditional machine learning approaches that depend on manually engineered features, deep learning models automatically learn hierarchical feature representations directly from raw image data. This capability allows them to identify subtle patterns, textures, and irregularities that may not be easily visible to the human eye. In medical imaging, deep learning has demonstrated remarkable success in tasks such as tumor detection, disease classification, and anomaly identification. Its ability to process large-scale datasets and continuously improve performance makes it highly suitable for developing reliable diagnostic solutions.

Among deep learning techniques, Convolutional Neural Networks (CNNs) have been the most widely used for image classification tasks. CNNs are specifically designed to capture spatial hierarchies in images by applying convolutional filters that extract local features such as edges, textures, and color variations. These networks operate through multiple layers, where each layer progressively learns more complex representations of the input image. Architectures such as EfficientNet have further enhanced CNN performance by introducing compound

scaling, which balances network depth, width, and resolution for optimal efficiency and accuracy. As a result, CNNs are highly effective in detecting fine-grained visual details required for distinguishing different types of skin lesions.

However, despite their success, CNNs exhibit inherent limitations when it comes to understanding global contextual information. Since CNNs operate using localized receptive fields, they primarily focus on small regions of an image at a time. This makes it challenging for them to capture long-range dependencies and overall structural patterns, such as asymmetry, border irregularity, and spatial distribution of features across the entire lesion. These global characteristics are essential in dermatological diagnosis, particularly when applying clinical rules such as the ABCD rule(Asymmetry, Border, Color, Diameter). As a result, relying solely on CNN-based models may lead to misclassification in complex cases where global context plays a significant role.

To overcome these limitations, Vision Transformers (ViTs) have recently emerged as a powerful alternative in computer vision tasks. Inspired by transformer architectures used in natural language processing, ViTs analyze images by dividing them into smaller patches and processing them as sequences of tokens. Through the use of self-attention mechanisms, Vision Transformers can model relationships between all parts of an image simultaneously. This enables them to capture global dependencies and understand the overall structure of visual data more effectively than traditional CNNs. Consequently, ViTs are particularly useful for tasks that require holistic analysis of images, such as identifying irregular lesion shapes and patterns.

Despite their advantages, Vision Transformers also have certain drawbacks. They generally require large amounts of training data and computational resources to achieve optimal performance. Additionally, since they do not inherently focus on local feature extraction, they may miss fine-grained details that are critical in medical image analysis. This creates a gap where neither CNNs nor Transformers alone can fully address the complexity of dermatological diagnosis.

To bridge this gap, the proposed system introduces a hybrid deep learning architecture that integrates the strengths of both CNNs and Vision Transformers. In

this approach, EfficientNet- B0 is used as the backbone network for extracting detailed local features from dermatoscopic images. These features are then passed through a 1×1 convolutional bridge, which transforms the feature maps into a format suitable for transformer processing. Subsequently, Vision Transformer blocks are employed to perform global reasoning by analyzing relationships across the entire image. This combination enables the model to simultaneously capture microscopic details and macroscopic structures, leading to improved diagnostic performance.

In addition to its core diagnostic capabilities, the system also emphasizes usability and real- time application. A web-based interface developed using Streamlit allows users to upload skin images and receive instant predictions along with confidence scores and medical recommendations. This not only enhances accessibility but also enables deployment in telemedicine environments, making healthcare services more inclusive and efficient.

Overall, this project demonstrates how hybrid deep learning architectures can significantly enhance the accuracy, reliability, and usability of computer-aided dermatological diagnosis systems. By combining local feature extraction with global contextual understanding, the proposed model addresses key limitations of existing approaches and contributes to the advancement of intelligent healthcare solutions.

In recent years, the integration of advanced deep learning architectures into healthcare systems has opened new possibilities for improving diagnostic accuracy and efficiency. The increasing availability of large annotated medical datasets, along with improvements in computational power, has enabled the development of highly sophisticated models capable of outperforming traditional methods. In dermatology, where visual assessment plays a critical role, such intelligent systems can act as reliable assistants by providing consistent and objective evaluations of skin lesions. This is particularly important in scenarios whereaccess to experienced dermatologists is limited.

Another significant aspect of modern diagnostic systems is their ability to generalize across diverse patient populations and imaging conditions. Skin lesion

images often vary due to differences in lighting, skin tone, imaging devices, and lesion characteristics. A robust deep learning model must therefore be capable of handling such variations while maintaining high performance. The hybrid architecture proposed in this work addresses this challenge by combining the robustness of convolutional feature extraction with the adaptability of attention-based mechanisms. This ensures that both fine-grained details and global structural relationships are effectively captured, leading to more reliable predictions.

Furthermore, the adoption of hybrid models represents a shift towards more biologically inspired computational approaches. Human experts typically analyze skin lesions by simultaneously observing local features, such as texture and pigmentation, and global characteristics, such as symmetry and border irregularity. By integrating CNNs and Vision Transformers, the proposed system mimics this dual-analysis process, thereby aligning machine perception more closely with human diagnostic reasoning. This alignment not only improves model interpretability but also enhances trust among medical professionals.
Scalability and real-time applicability are also critical considerations in the design of modern healthcare systems. The proposed model is optimized to balance accuracy and computational efficiency, making it suitable for deployment in real-time environments such as web applications and mobile platforms.

The inclusion of an interactive user interface further enhances the system's usability by allowing seamless interaction between users and the underlying AI model. This ensures that the technology is not confined to research settings but can be effectively translated into practical healthcare solutions.

1.2 Problem Statement

One of the main issues with automated dermatology: Visual likeness among classes It's very difficult to distinguish between many types of skin lesion, because visually an benign seborrheic keratosis and a cancerous melanoma may appear almost identical. Standard computer-aided diagnosis (CAD) models have so far typically been built around single Convolutional Neural Network (CNN).

As opposed to being able to interpret the “global” layout of a skin lesion, CNNs “see” small local sections of images. Thus they “take” the textures, colors and some pigment patterns into account, but lack the “receptive field”.

In the clinic, dermatologists utilize the ABCD rule (Asymmetry, Border irregularity, Color variation, and Diameter) to distinguish malignant skin from benign one. Local features alone would fail to detect asymmetric distribution over the whole image or irregular border that appears across the whole lesion. TheHAM10000dataset itself has highly imbalanced classes with significant disparity between healthy-to-malignant (benign disease are highly frequent compared to rare but fatal malignant ones).

Typically trained on this kind of data, existing deep learning models suffer from the majority class-bias which makes them have high overall performance yet alarmingly poor detection performance of rare and clinically significant malignant patterns. Thus, there’s a pressing need of an effective hybrid reasoning system which has the following capability: (1) Simultaneously leverage both local texture and global contextual cues; (2) Sustain performance on the minority class with balanced dataset; (3) Offer clear and immediate interaction which allows the complicated deep network prediction to be translated into useful medical indications for care.

1.3 Objectives of the Study

The primary objective of this study is to design and develop an advanced computer-aided diagnostic system for accurate classification of skin diseases using a hybrid deep learning approach. The study focuses on leveraging the strengths of both convolutional neural networks and transformer-based architectures to overcome the limitations of traditional methods.

By integrating EfficientNet-B0 and Vision Transformer (ViT), the system aims to provide a more comprehensive understanding of dermatoscopic images by capturing both local and global features, thereby improving diagnostic reliability and performance.

1. To design and develop a hybrid deep learning architecture

The foremost objective of this study is to construct a novel hybrid model that

combines EfficientNet-B0 for local feature extraction and Vision Transformer for global feature analysis. This integration is achieved using a 1×1 convolutional bridge that ensures smooth transition between convolutional feature maps and transformer tokens. The goal is to create a unified architecture capable of performing both microscopic and macroscopic analysis of skin lesions within a single framework.

2. To enhance classification accuracy and reliability

Another key objective is to improve the overall accuracy and robustness of skin disease classification. Traditional standalone models often fail to generalize across diverse lesion types due to their limited feature representation capabilities. This study aims to reduce misclassification by enabling the model to learn richer and more discriminative features, thereby achieving higher accuracy and consistency across all seven classes of the dataset

3. To capture both local and global image features

A significant objective of this work is to ensure that the model effectively captures:

Local features such as texture, pigmentation, and fine patterns

Global features such as asymmetry, border irregularity, and lesion structure

By combining these two perspectives, the system mimics the diagnostic approach used by dermatologists, leading to more accurate and clinically relevant predictions.

4. To address class imbalance in the dataset

Medical datasets such as HAM10000 often suffer from severe class imbalance, where certain classes dominate the dataset while others are underrepresented. This objective focuses on implementing techniques such as weighted random sampling and data augmentation to ensure balanced learning.By doing so, the model can maintain high sensitivity for rare but critical conditions like melanoma, thereby improving fairness and reliability.

5. To develop a real-time diagnostic web-based system

The study also aims to design and implement an interactive web-based interface using modern frameworks such as Streamlit. This interface allows users to upload skin images and receive instant diagnostic predictions. The goal is to make the system accessible, user-friendly, and suitable for real-time applications in both clinical and non-clinical environments.

6. To provide automated health recommendations and insights

Beyond classification, the system aims to generate meaningful outputs that include confidence scores, risk levels, and basic healthcare recommendations. This objective ensures that the system not only predicts disease categories but also assists users in understanding the results and taking appropriate actions, thereby increasing its practical utility.

7. To evaluate model performance using comprehensive metrics

A critical objective of this study is to rigorously evaluate the performance of the proposed model using standard evaluation metrics such as accuracy, precision, recall, and Macro- averaged F1-score. These metrics ensure that the model performs consistently across all classes, especially in imbalanced datasets, and provides a fair assessment of its effectiveness.

8. To ensure scalability and real-world applicability

The final objective is to design a scalable and adaptable system that can be extended for real- world deployment. The system should be capable of integration into healthcare platforms such as hospital systems, telemedicine applications, and mobile-based diagnostic tools. This ensures that the proposed solution is not limited to theoretical research but has practical relevance in improving healthcare delivery.

2

LITERATURE REVIEW

LITERATURE REVIEW

[1] **Philipp Tschandl, Cliff Rosendahl, and Harald Kittler (2018). The HAM10000 Dataset.**

This paper introduces the HAM10000 dataset, a large collection of dermatoscopic images used for training and evaluating skin lesion classification models. The dataset includes seven categories of skin diseases and is widely used in deep learning research. However, it presents a significant challenge due to class imbalance, where certain classes dominate the dataset. This makes it difficult for models to accurately detect rare but critical conditions such as melanoma.

[2] **Mingxing Tan and Quoc V. Le (2019). EfficientNet: Rethinking Model Scaling**

This study presents EfficientNet, a family of convolutional neural networks that improve performance through compound scaling of depth, width, and resolution. EfficientNet achieves high accuracy with fewer parameters, making it suitable for medical image analysis. However, like other CNN-based models, it primarily focuses on local feature extraction and lacks the ability to capture global contextual relationships.

[3] **Alexey Dosovitskiy et al. (2021). Vision Transformers for Image Recognition.**

This paper introduces Vision Transformers (ViT), which apply transformer-based architectures to image classification tasks. By dividing images into patches and using self-attention mechanisms, ViT captures global dependencies across the image. While this approach improves understanding of overall structure, it requires large datasets and may miss fine-grained local details important in medical diagnosis.

[4] **Andre Esteva et al. (2017). Dermatologist-Level Classification of Skin Cancer .**

This research demonstrates that deep learning models can achieve dermatologist-level accuracy in skin cancer classification. The study uses large-scale datasets and CNN architectures to classify skin lesions effectively.It highlights the potential of AI in dermatology but also shows

limitations in generalization and reliance on local features.

[5] **Kaiming He et al. (2016). Deep Residual Learning for Image Recognition.**

ResNet introduces residual connections that enable the training of very deep neural networks. It has been widely used in medical image classification tasks, including skin disease detection. Despite its success, the model is limited by its focus on local features and inability to fully capture global structural patterns.

[6] **Ze Liu et al. (2021). Swin Transformer: Hierarchical Vision Transformer.**

This paper proposes the Swin Transformer, which introduces a hierarchical structure and shifted window attention mechanism. It improves efficiency and captures both local and global features. However, it still requires high computational resources and does not fully address fine-grained feature extraction challenges in medical imaging.

[7] **Yuan Liu et al. (2020). A Deep Learning System for Differential Diagnosis**

This study presents a deep learning system capable of diagnosing multiple skin diseases with high accuracy. It demonstrates the practical application of AI in clinical settings. However, the system relies heavily on CNN-based architectures and faces challenges in handling class imbalance and global feature understanding.

[8] **S. Haque, S. Rahman, and S. M. M. Rahman (2023). Hybrid CNN and Vision Transformer for Skin Lesion Classification. Biomedical Signal Processing andControl.**

This research proposes a hybrid model combining CNN and Vision Transformer architectures for skin lesion classification. The modelleverages CNNs for local feature extraction and transformers for global context understanding. The results show improved accuracy and robustness compared to standalone models, highlighting the effectiveness of hybrid approaches in medical image analysis.

3

SYSTEM ANALYSIS

SYSTEM ANALYSIS

3.1 Software development lifecycle

There are various software development approaches defined and designed which are used/employed during development process of software, these approaches are also referred as "Software Development Process Models". Each process model follows a particular life cycle in order to ensure success in process of software development. Fig 3.1 Software Development Life Cycle

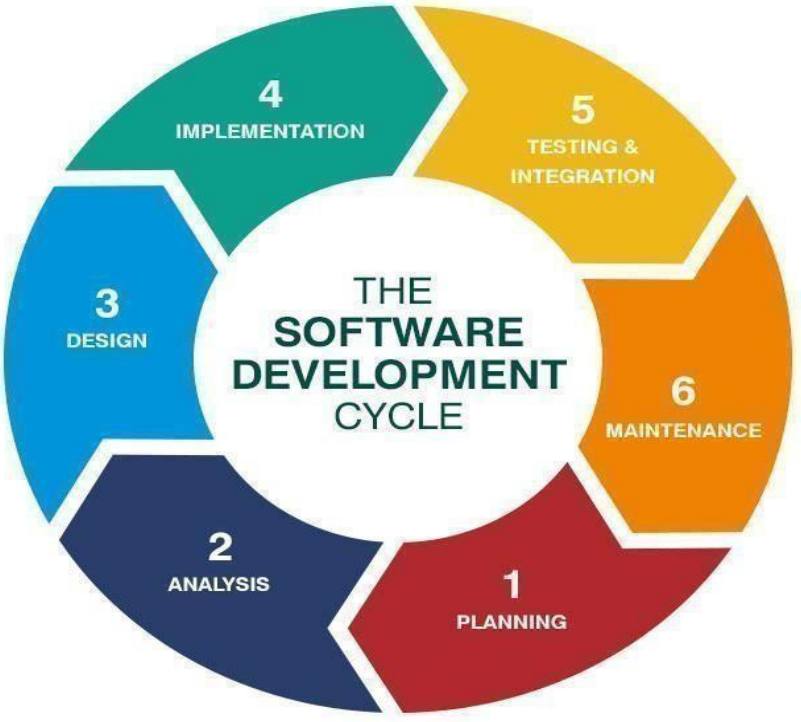


Fig 3.1 Software development lifecycle

3.2 Overview of Deep Learning in Healthcare

An integral piece of what drives our health information future is Deep Learning (DL), which goes beyond basic automation to serve as high-level cognitive assistance in our clinical decision-making processes. As the application of multi-tiered artificially structured neural networks, DL has the unique ability to auto-program itself and learn to recognise extremely complex biological structures from massive datasets such as the 10,015 images in our HAM10000 dataset, entirely unprompted. For clinical purposes, this results in algorithms capable of

recognising 'hidden' biomarkers buried in a photograph of a mole, a radiology image, or even genomic sequences that the unaided eye could fail to spot after too much use.Medical adoption is increasingly centered on using DL's ability to interpret and glean insights from unstructured data that dominate the realm of medical files -from pixel data in a dermatoscopic image or time-series data in an EKG tracing -DL systems like the hybrid one designed in this article can deliver an objective and reproducible secondary opinion without ever getting tired, tired by eye strain or distracted by personal fatigue.

Applications of AI/DL in Healthcare

The versatility of deep learning allows it to be applied across several critical domains of medicine:

- **Medical Image Analysis:** This is the most prominent application, where models like CNNs and Transformers are used for the classification of skin cancer, detection of tumors in MRI/CT scans, and identifying retinal diseases through fundus photography.
- **Predictive Diagnostics:** DL algorithms analyze electronic health records (EHR) to predict patient outcomes, such as the likelihood of heart failure, sepsis onset, or the progression of chronic diseases like diabetes.
- **Drug Discovery and Genomics:** AI accelerates the identification of potential drug candidates by simulating molecular interactions and mapping genetic sequences to identify mutations responsible for rare hereditary conditions.
- **Virtual Health Assistants:** Real-time interfaces, such as the Streamlit-based system developed in this project, allow for immediate preliminary triage, providing patients with risk assessments and health advice before they even reach a specialist.
- **Pathology and Histology:** Deep learning automates the analysis of tissue slides, assisting pathologists in grading cancer stages with higher precision and speed than manual microscopic inspection.

3.3 Dataset Analysis: The HAM10000 Dataset

The HAM10000 ("Human Against Machine with 10,000 training images") is one of the largest available benchmark datasets in digital dermatology. Due to the relatively small size and limited diversity in prior datasets it was shared as the "10,000 training images for humanity against machine for training and validation" to train machine deep learning models. This dataset contains a total of 10,015 dermatoscopic images of different
patients acquired from a diverse background using various dermatoscopes.The dataset is categorized into seven distinct diagnostic classes, representing a broad spectrum of common and rare skin conditions.

Fig 3.3 Classes of HAM10000 Dataset

These include:

- **Melanocytic Nevi (NV):** Benign proliferations of melanocytes (moles).
- **Melanoma (MEL):** A malignant neoplasm derived from melanocytes; the most lethal form of skin cancer.
- **Benign Keratosis-like Lesions (BKL):** Including seborrheic keratoses and solar lentigines.
- **Basal Cell Carcinoma (BCC):** A common, locally invasive malignant skin neoplasm.
- **Actinic Keratoses (AKIEC):** Pre-cancerous lesions that can progress to squamous cell carcinoma.
- **Vascular Lesions (VASC):** Including angiomas and pyogenic granulomas.
- **Dermatofibroma (DF):** A benign skin tumor.

From a technical stand point this data set has been a very good challenge to tackle in that Class Imbalance has been very extreme within the total dataset. Roughly 67% of the images belong to class NV (Nevi) whereas DF and VASC were below 2% of the total. A normal Neural Network would essentially get ‘stuck’ classifying all the images as NV without techniques of over-sampling, of which the Weighted Random Sampler, as was used in this project, was identified. It was through analysing these distribution pattern which the system analysis phase used, to check sensitivity for these malignant cases had not been sacrifice

3.4 Handling Class Imbalance in Lesion Categories

The very high level of imbalance in the HAM10000 Dataset creates a significant challenge for classifying the data. To illustrate this: the largest class - ‘Melanocytic Nevi’ (NV) - makes up >65% of the entire data set and proportional counts for lower frequency classes such as ‘Melanoma’ (MEL), ‘Dermatofibroma’ (DF), or ‘Vascular lesions’ (VASC) are low (hence they are under-represented). If modelling occurs without addressing this class imbalance, the resulting model will contain a majority class bias (i.e., the model will achieve high total accuracy regardless of the model's ability to predict clinically relevant lesion types).

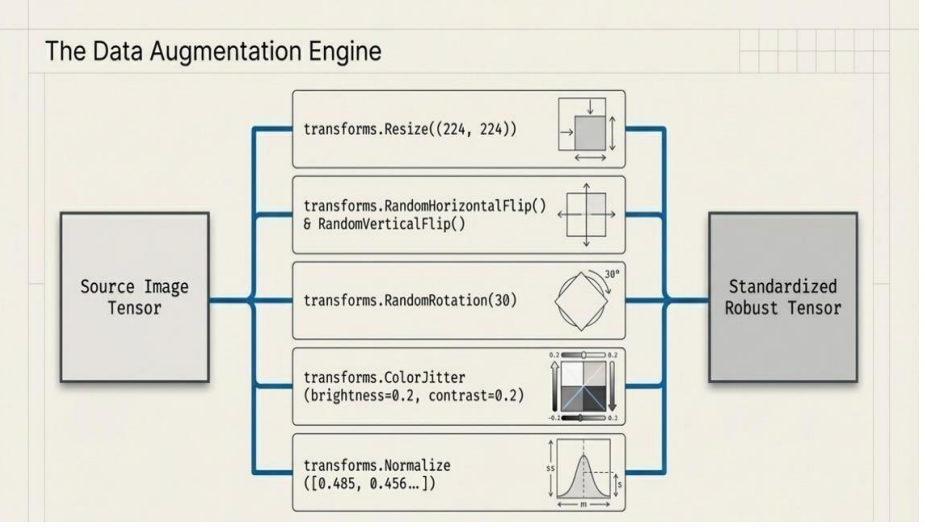


Fig 3.2 Augmentation Engine

Using Weighted Random Sampling offers a means to mitigate this class imbalance. By assigning under-represented classes a higher sampling weight at data load time, the number of examples of the rare versus more prevalent classes used during model training will be closer to that of the common classes than to that of the under-represented class.The change in data collection method will allow better model learning of the features across all classes (including both dominant and rare class), and the resulting increase in sensitivity and F1 macro-averaged performance metrics across all categories will demonstrate significant improvements in each of the seven types of lesions

3.5 Layered Architecture of the Hybrid Model

The suggested framework employs a multi-layered hybrid architecture to realise a "Local- Global Diagnostic Reasoning". This architecture is envisioned to simultaneously extract microscopic textural information of a lesion in skin image and its macroscopic architectural features, such as border uniformity, general symmetry. The architecture can be divided into 4 main functional stages that operate in a pipelines order.

1. Local Feature Extraction Layer (CNN Backbone)

The first layer of the architecture is an efficientnet-B0 used as a convolutional backbone layer. This convolutional layer treats raw pixels of the dermatoscopy image as input for extracting low and mid level features. Due to the efficient use of inverted bottleneck mobile convolution (MBConv) in the EfficientNet-B0, the layer is very efficient in extracting local patterns likepigment network, streaks, and dots and generate a set of high dimension feature maps representing the spatial property of the lesion.

2. Dimensional Alignment Layer (1x1 Convolutional Bridge)

Another major obstacle with hybrid models is the ‘representation gap between CNN and Transformers. CNN models output feature maps that are 3D (height x width x channel) while a Transformer model takes in 1D sequences of tokens as input. In

order to bridge this gap, we've added a 1x1 Convolutional Bridge. This 1x1 Convolution reduces the number of channels and makes a 2D grid of information into a sequence of "visual tokens" that the attention mechanism can process while ideally losing little valuable spatial information.

3. Global Reasoning Layer (Vision Transformer Blocks)

The flattened image data is then fed into a sequence of Transformer Blocks. Instead of relying on CNNs' pre-programmed kernels, Transformer blocks are designed to make use of Self-Attention models. Each pixel of the "lesion" has the ability to pay attention to all the other "pixels." By this mechanism, the model can compute dependencies across long distances. This stage carries out the global reasoning task to ascertain whether a bumpy border of a lesion on one part has relevance in pattern of abnormality across from it. In essence, this is very akin to the way that specialists make clinical assessment of the skin lesion.

4. Classification and Decision Layer

Last we have the Global Average Pooling (GAP) and then we have a Fully Connected (FC) head that would finally give us the probability distribution across the seven classes. These local and global level feature representation combined to get us a final probability distribution across the 7 disease classes in form of Softmax activated prediction with Confidence scores.

3.6 Features of EfficientNet and Vision Transformers

Two of the leading architectures in contemporary Computer Vision are EfficientNet and Vision Transformer, and together these complementary architectures can lead to improved accuracy in skin disease classification applications. EfficientNet excels at local feature extraction - such as extracting texture, lesion boundaries, and colour changes from a lesion. In addition, EfficientNet uses a technique known as compound scaling, which allows for simultaneous simultaneous optimisation of depth, width, and resolution and also provides the ability to learn detailed and nuanced visual patterns while being

computationally efficient; from a computational perspective, this makes EfficientNet very well-suited for performing medical image analysis.

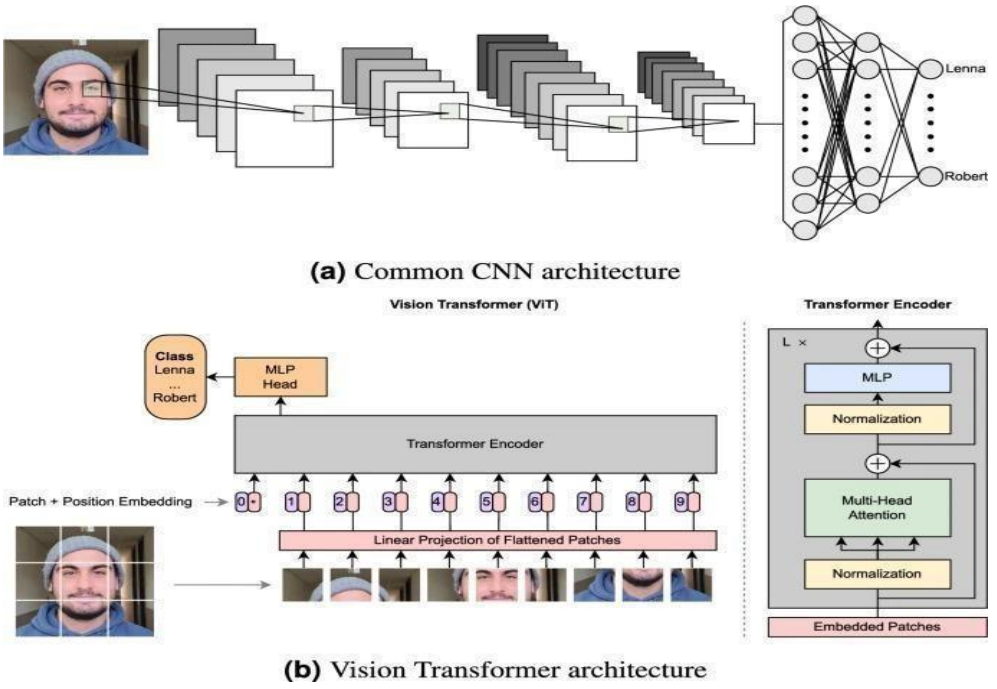


Fig 3.6 Comparison of CNN&ViT architectures

MBConv layers and squeeze and excitation of EfficientNet are used to help with the optimization and identify the most relevant parts of an image while reducing the number of unnecessary computations. By doing this, it is able to achieve both a high level of accuracy and a low level of computation (making it ideal for use in real-time healthcare applications). Its ability to identify very finely detailed localized regions helps it to differentiate between closely related skin diseases accurately. Unlike previous methods that focused on analyzing images locally, Vision Transformers break down images into smaller patches and apply self-attention mechanisms globally to analyze them. This allows the entire image to be analyzed for relationships, including spatial distribution and symmetry. EfficientNet utilizes local feature extraction, while the transformer captures global context, thereby resulting in a hybrid model that both improves diagnostic accuracy and reduces misclassification.

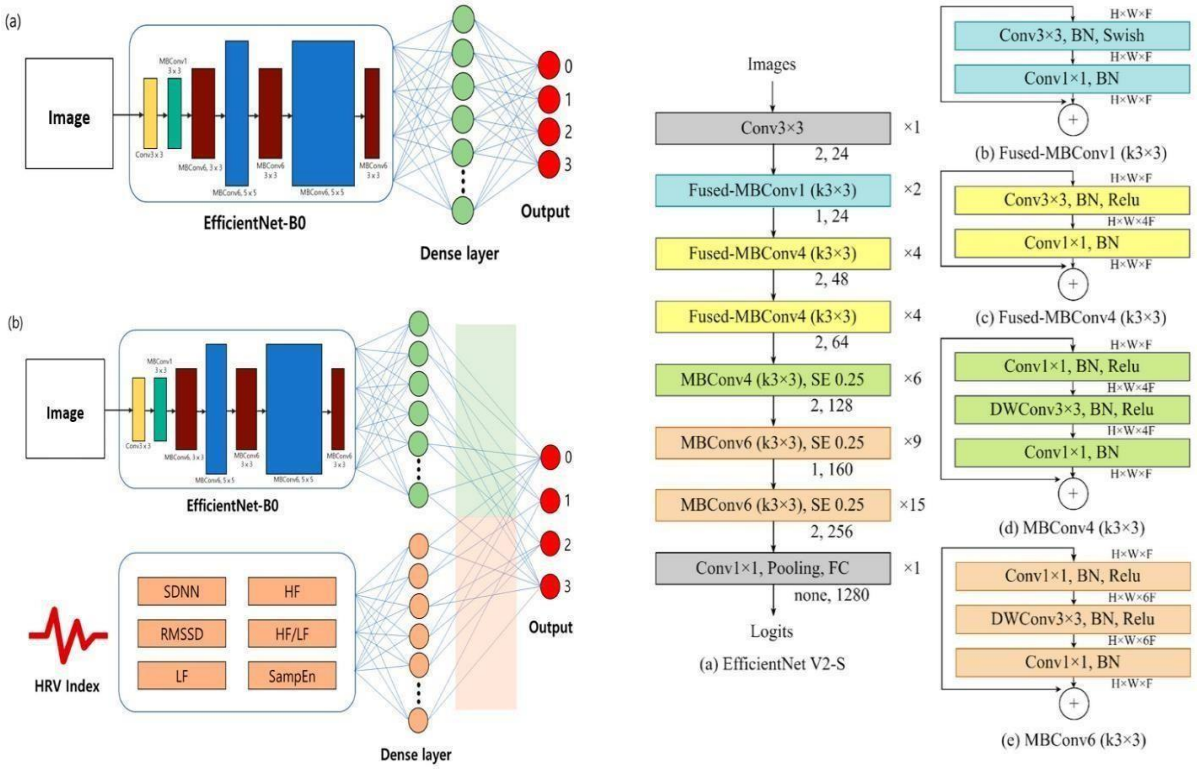


Fig 3.6 Working of EfficientNet-B0

Merging EfficientNet + Vision Transformer allows for effective local feature extraction along with an effective understanding of the entire image globally. The combined approaches will allow greater reliability in overall diagnosis and improve accuracy in classifying different skin diseases.

3.7 Functions of the Diagnostic System

Dermatological images are processed by an advanced diagnostic system through a structured workflow from the input of the raw image to the output of the diagnostic classification. The image is acquired through a web-based format before being resized and standardized (via normalization and noise removal) as part of the preprocessing and preparation stages of the diagnostic pipeline. After preprocessing has been completed, EfficientNet will be used to extract detailed local-level features from the image being analyzed, the details of the local features will be passed to a Vision Transformer using a convolutional bridge, where the

vision transformer will use the information from the local feature extraction and determine how to classify the image into the appropriate disease categories based off of their confidence levels, which will be accompanied with an interpretable user-friendly report as well as any medically relevant information regarding each identified skin disease. Therefore, the complete end-to-end workflow allows for the real-time diagnostic assessment of skin diseases .

3.8 Applications of AI in Dermatology

- Skin Disease Classification and Diagnosis
- Early Detection of Skin Cancer
- Image Segmentation and Lesion Analysis
- Severity Assessment and Disease Monitoring
- Teledermatology and Remote Diagnosis
- Personalized Treatment Recommendations
- Automated Report Generation
- Mobile-Based Dermatology Applications
- Clinical Decision Support Systems

3.9 Existing Methods

3.9.1 Traditional Clinical Diagnosis (Visual Inspection)

Visual examination by physicians is the basis for diagnosing skin disorders based on clinical observation performed by experienced physicians in their analysis of the color, shape, size and distribution of lesions on the skin. Dermatologists determine a diagnosis based on their experience and education and therefore can make an accurate diagnostic determination; this however has the disadvantage of some subjectivity, dependance upon physician training and experience and restricted numbers of physicians experienced in diagnosing skin disease in rural locations, as well as the possibility for observing fatigue resulting in incorrect identification of skin lesions and for multiple observers to incorrectly interpret skin lesions. In addition, to perform the manual diagnosis, a considerable amount of time must be devoted; therefore screening large populations will be more

difficult to perform. All of these limitations identically demonstrate a need to develop automated assistance systems.

3.9.2 Standalone CNN Models (Ex:EfficientB0)

A standalone CNN is a deep learning architecture designed to process data with a grid-like topology (primarily images). Its "standalone" nature refers to its ability to perform **end-to-end learning**: taking raw pixels as input and outputting a classification or detection without requiring manual feature engineering.

Key Structural Components

To explain how these models work in your report, focus on these three pillars:

1. **Convolutional Layers:** These act as "filters" that scan the image to detect patterns like edges, textures, and eventually complex shapes.
2. **Pooling Layers (Downsampling):** These reduce the spatial dimensions of the data, making the model more computationally efficient and helping it focus on the most important features (e.g., **Max Pooling**).
3. **Fully Connected (FC) Layers:** Located at the end of the network, these layers take the high-level features extracted by the convolutions and use them to classify the image into a specific category.

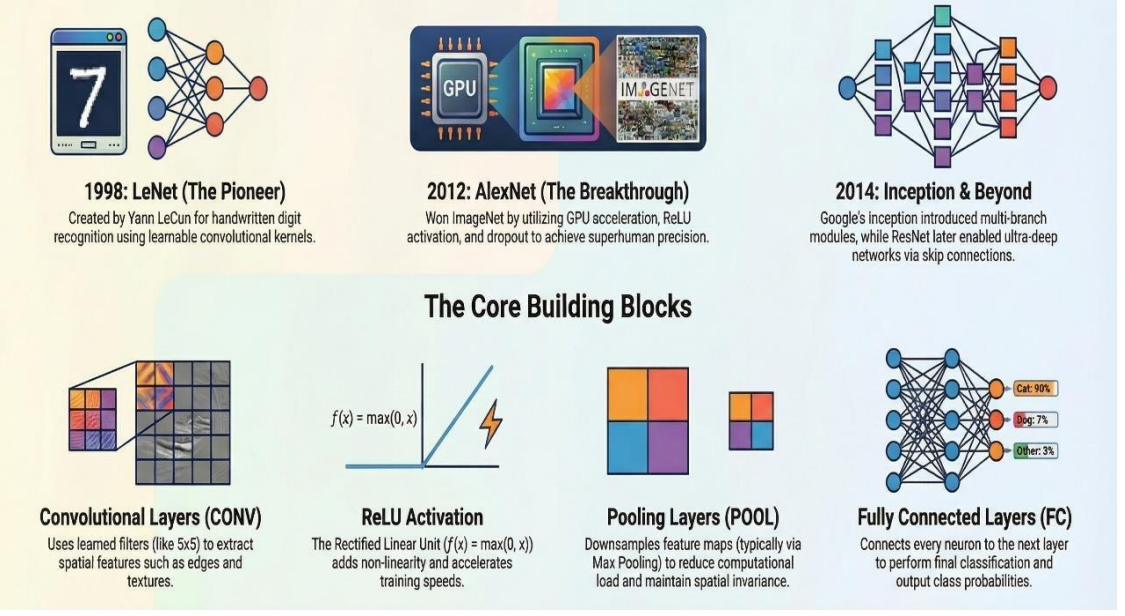


Fig 3.9.2 Standalone CNN architecture

3.9.3 Standard Vision Transformers

Vision Transformers introduced attention-based learning into computer vision tasks. These models analyze entire images by understanding relationships between patches.They perform well in capturing global structures but require large datasets and computational resources for effective training. Without strong local feature extraction, transformers may miss fine texture information essential for dermatological analysis.Therefore, using transformers alone may not fully capture microscopic lesion details.

3.10 Proposed Methods

The proposed system addresses limitations of existing approaches by combining convolutional and transformer-based learning into a unified framework.

3.10.1 Hybrid Local-Global Reasoning

EfficientNet is used as the local feature extraction component of the hybrid architecture, while the Vision Transformer block serves as the global reasoning component of the hybrid architecture. EfficientNet extracts fine-grained characteristics of lesions, whereas the transformer layer examines spatial relationships across the entire image. The 1 × 1 convolutional bridge aligns feature channels prior to the transformer being used to process them, and this will ensure there is a smooth transition from one architecture to another in terms of the flow of information. By creating this form of hybrid reasoning, we improved the accuracy of classification through the combination of fine-grained detail and higher-level context.The system therefore learns both microscopic and macroscopic disease patterns simultaneously, resulting in robust diagnostic performance.

3.10.2 Real-Time Diagnostic Web Interface

This program encompasses an interactive web interface that provides a means of direct interaction between the user and an artificial intelligence (AI) model. Users upload an image of their skin via an internet browser, and the backend uses the

trained hybrid model to process it.Once complete, the predicted type of disease, the percentage of confidence associated with that prediction, and some advice on what to do next to manage one's health are all displayed to the user on the web interface.To ensure quick response time, real time processing techniques are employed so that this platform would be valid for telehealth/remote screening uses.

3.10.3 Automated Health Advisory System

Beyond prediction, the system generates automated health recommendations based on diagnosis results. After classification, predefined medical guidance modules provide preventive suggestions, lifestyle advice, and indications for consulting specialists

.This advisory component enhances usability by transforming predictions into actionable insights. It helps users understand next steps rather than receiving only technical outputs.The advisory system also promotes awareness and encourages early medical consultation, improving healthcare outcomes.

4

SYSTEM REQUIREMENT SPECIFICATIONS

SYSTEM REQUIREMENT SPECIFICATIONS

System Requirement Specifications define the technological foundation required for designing, implementing, and deploying the proposed hybrid deep learning system for skin disease classification. These specifications act as a structured guideline describing the computing environment, supporting software, and operational conditions necessary for efficient system execution.

4.1 Software Requirements

Software requirements refer to the collection of programming environments, development tools, and computational libraries necessary for implementing the hybrid diagnostic model. These requirements are selected based on performance efficiency, compatibility with deep learning frameworks, and ease of integration with visualization platforms.

The project relies on a specific stack of Python libraries: Core Framework: PyTorch and torchvision for model building, training, and image transformations

- **Model Components**: The timm (PyTorch Image Models) library is required to instantiate the EfficientNet-B0 backbone and the Vision Transformer (ViT) blocks
- **Data Handling:** Pandas for metadata CSV processing and PIL (Pillow) for image loading and conversion
- **Evaluation & Visualization**: scikit-learn for calculating metrics (F1-score, Recall, etc.), and Matplotlib/Seaborn for generating the confusion matrix and performance charts
- **Utilities:** tqdm for progress tracking during training

5

SOFTWARE ENVIRONMENT

SOFTWARE ENVIRONMENT

The software environment will be the integrated platform in which the hybrid deep learning model is developed, trained, tested, and deployed. It consists of a collection of programming languages, development tools, and machine learning frameworks, that form an integrated workflow, allowing the project's entire life cycle to happen within this environment.A well- developed software environment increases productivity because it allows for easier experimentation and reduces the complexity of development. By having a unified software environment for implementing algorithms, visualizing results, and evaluating performance, researchers will have a better overall experience.

The selected environment has been designed with flexibility and scalability in mind. With the rapid evolution of artificial intelligence research, it is imperative that the software stack be designed so that researchers can easily add new algorithms and/or datasets.The selected technologies will enable researchers to develop software using a modular approach, with the ability to change or replace preprocessing, training, and deployment components independently from each other.The software environment also provides opportunities for collaboration between researchers. By providing standardized tools, multiple developers can work on different components of the software simultaneously, without any compatibility issues. The use of version control will allow researchers to systematically track changes to a project throughout its lifetime.In addition, the software environment provides cross-platform compatibility for the software developed. This means that the same software can be run on several different computer platforms (e.g., Windows and Mac) with little or no change in the code, thus improving the portability and usability of the software produced.

5.1 Python Programming in VS Code

Python serves as the primary programming language for system implementation due to its simplicity, flexibility, and extensive scientific computing ecosystem. Its readable syntax enables rapid development and simplifies debugging, making it ideal for resear ch-based projects.Python

provides powerful libraries that support essential operations required for this project:These libraries reduce implementation complexity by offering optimized built-in functions for mathematical operations and data manipulation. Visual Studio Code (VS Code) is used as the development environment because it provides an efficient coding workspace with advanced productivity features. The editor supports intelligent auto-completion, error highlighting, integrated debugging, and customizable extensions.VS Code enables developers to manage virtual environments directly within the interface. This ensures controlled dependency management and prevents conflicts between library versions. The integrated terminal allows execution of training scripts and monitoring of outputs without switching applications.

5.2 Deep Learning Framework: PyTorch

The primary framework used for Deep Learning today is PyTorch, which benefits from flexibility and performance and is widely adopted by the research community. It contains all of the tools needed to create, train and evaluate neural networks without too much complexity involved.A major strength of PyTorch is its Dynamic Computation Graph, allowing you to modify the structure of your model at runtime. This is a tremendous advantage for working with hybrid architectures that combine both Convolutional Networks and Transformers during experimentation. PyTorch represents your data as Tensors, which allows for fast mathematical processing using optimized procedures for Parallel Computing. It also has Automatic Differentiation built-in, making iteasy to compute gradients for implementing the Backpropagation Algorithm that types of Neural Networks require.Transfer Learning is another significant advantage of using PyTorch. By Importing Pre-trained Efficient Nets, for example, you can use them to classify images using very little time and effort while improving accuracy.

6

SYSTEM DESIGN

SYSTEM DESIGN

In this chapter, you will learn about how the EfficientNet-B0 and Vision Transformer architectures can be combined into one hybrid model to assist physicians in making more accurate diagnoses of skin lesions.

To accomplish this, we created an image processing system that can efficiently classify dermatoscopic images by extracting features from Convolutional Neural Networks (CNNs) and providing contextual reasoning using a vision transformer. As a result, the architecture will automatically analyze both the fine-grain cellular details of each lesion and the overall structure of the lesion simultaneously, which will enhance the reliability of the physician's final diagnostic determination.The system design consists of three major components:

1. Local Feature Extraction using EfficientNet-B0

2. Feature Transformation through a 1×1 Convolutional Bridge

3. Global Context Modeling using Vision Transformer Blocks

Together, these components form a unified hybrid architecture capable of learning complex dermatological patterns.

6.1 Hybrid Architecture

The hybrid architecture integrates Convolutional Neural Networks (CNNs) and Transformer- based learning mechanisms into a single framework. Traditional CNNs are effective at extracting local spatial features but lack the ability to capture long-range dependencies across images. Conversely, Vision Transformers excel at modeling global relationships but require structured feature representations as input. To overcome these limitations, the proposed system combines EfficientNet- B0 for local feature extraction with Vision Transformer blocks for global reasoning. A bridging mechanism is introduced to ensure compatibility between convolutional outputs and transformer inputs. This layered approach enables the model to understand both microscopic and macroscopic lesion characteristics.

6.1.1 Local Feature Extraction

Local feature extraction is performed using the EfficientNet-B0 architecture, which serves as the backbone network of the system. EfficientNet is selected due to its optimized scaling strategy that balances network depth, width, and resolution, allowing efficient computation without sacrificing performance.

In dermatological image analysis, local features such as pigmentation patterns, texture irregularities, color variations, and cellular abnormalities play a critical role in diagnosis. EfficientNet-B0 captures these details through stacked convolutional layers and mobile inverted bottleneck convolution (MBConv) blocks.

Working Principle

The working principle of the Hybrid Skin Lesion Classification Project is based on a synergistic deep learning pipeline that combines local feature extraction with global contextual understanding to diagnose seven distinct skin conditions The system operates through four primary phases:

1. Data Processing and Class Balancing

The workflow begins by loading the HAM10000 dataset, which is often highly imbalanced

Preprocessing: Images are resized to 224x224 pixels and normalized using ImageNet statistics During training, data augmentation (flips, 30° rotations, and color jitter) is applied to prevent overfitting Addressing Imbalance: To ensure the model learns rare conditions effectively, a WeightedRandomSampler assigns weights to each image based on the inverse frequency of its class (1/count). This ensures that minority classes are represented just as often as common ones in each training batch

2. Hybrid Feature Extraction Architecture

The core innovation lies in the HybridSkinModel, which merges two different types of neural networks to maximize diagnostic accuracy : Local Spatial Extraction

(CNN): An EfficientNet-B0 backbone extracts fine-grained local features (like borders and textures) from the skin lesion. It produces a high-dimensional feature map with 320 channels

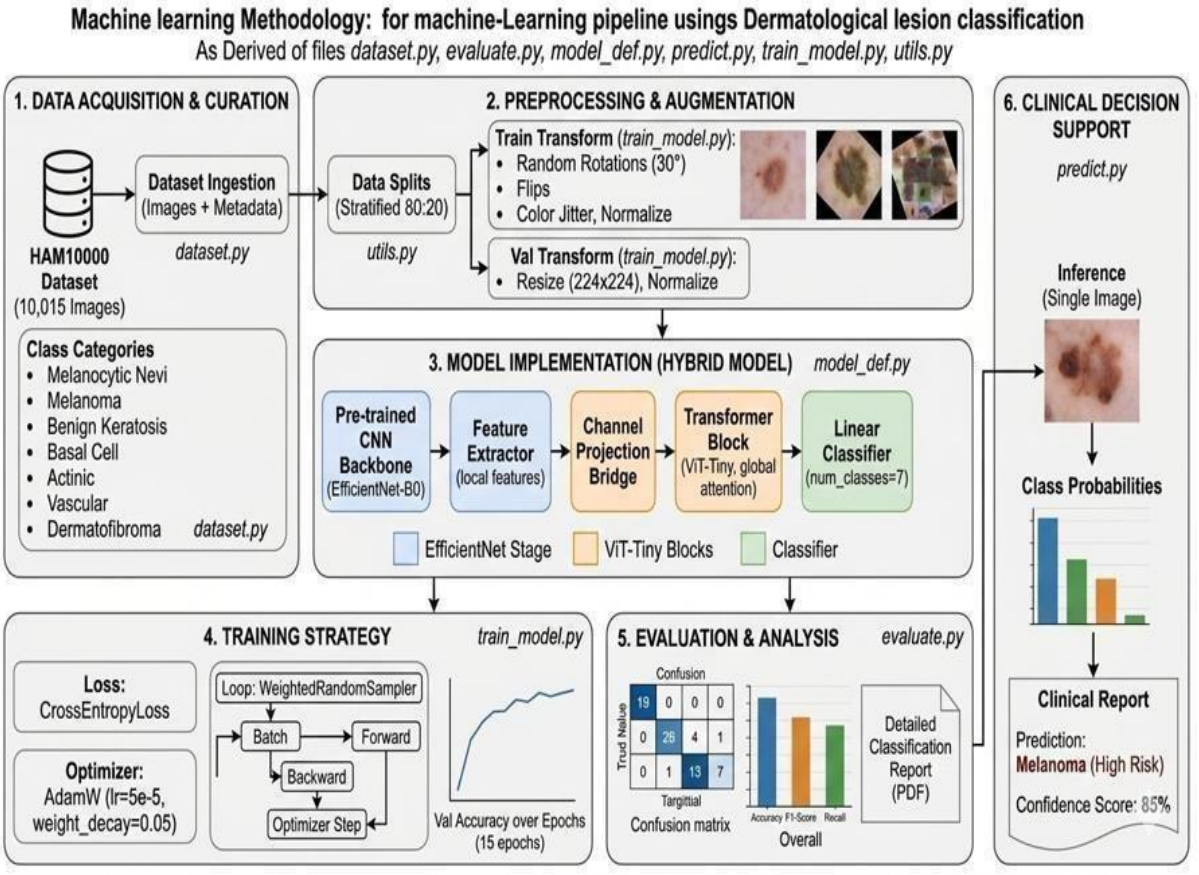


Fig 6.1.1 Hybrid Architecture of Efficientnet-B0 + ViT

The Bridge Layer: A 1x1 convolution layer acts as a bridge, reducing the 320 channels to 192 channels to align with the input requirements of the transformer . Global Contextual Modeling (ViT): The spatial features are flattened into tokens and passed through Vision Transformer (ViT) blocks.

Unlike standard CNNs, these blocks use self-attention to understand the long-range dependencies and global context across the entire lesion image

4. Diagnostic Inference and Evaluation

Metric Rigor: The system is evaluated using Macro F1-score, Recall, and Precision to provide a balanced view of performance across all seven classes, rather than relying solely on raw accuracy Clinical Prediction: For real-world use, the Predictor class loads the saved model and applies a Softmax function to the raw output. This converts the data into a confidence percentage for each disease type, which is then mapped from technical codes (e.g., 'mel') to human-readable clinical names like "Melanoma".

6.1.2 1×1 Convolutional Bridge and Channel Alignment

The outputs produced by convolutional networks differ structurally from the inputs required by transformer models. Vision Transformers operate on sequential embeddings rather than spatial feature maps. Therefore, a transformation mechanism is required to bridge this architectural gap.The proposed system introduces a **1×1 convolutional bridge**, which converts convolutional feature maps into transformer-compatible embeddings.

Purpose of the 1×1 Convolution

The 1×1 convolution performs the following functions:

- Adjusts channel dimensions
- Preserves spatial information
- Reduces computational complexity
- Converts features into latent representations

Unlike larger convolution kernels, a 1×1 convolution operates across channels without altering spatial resolution. This enables efficient feature projection while maintaining essential information extracted by EfficientNet.

6.1.2 Global Reasoning (Vision Transformer Blocks)

After feature transformation, the embeddings are passed into Vision Transformer (ViT) blocks for global reasoning. Unlike CNNs, which analyze images using

localized receptive fields, transformers evaluate relationships between all image regions simultaneously through attention mechanisms.

Self-Attention Mechanism

The Vision Transformer uses multi-head self-attention to determine how different parts of the image relate to one another. Each image patch interacts with every other patch, allowing the model to capture global dependencies.

For dermatological diagnosis, this enables analysis of:

- Lesion symmetry
- Shape irregularities
- Border distribution
- Overall spatial arrangement

Transformer Block Components

Each transformer block consists of:

- Multi-head self-attention layer
- Layer normalization
- Feed-forward neural network
- Residual connections

Benefits of Global Reasoning

- Captures long-range dependencies
- Improves contextual understanding
- Enhances classification robustness
- Reduces feature bias toward local regions

The transformer output represents a comprehensive understanding of the lesion combining both global and local insights

6.2 Block Diagram

Introduction to the Block Diagram

The block diagram represents the structural workflow of the proposed hybrid diagnostic system. It visually explains how data flows through different components from input acquisition to final prediction. The diagram provides a simplified representation of the internal working mechanism of the system by dividing the entire diagnostic process into interconnected functional modules. Each block performs a specific operation, and together they form an intelligent pipeline capable of analyzing dermatoscopic images and generating accurate diagnostic outcomes.

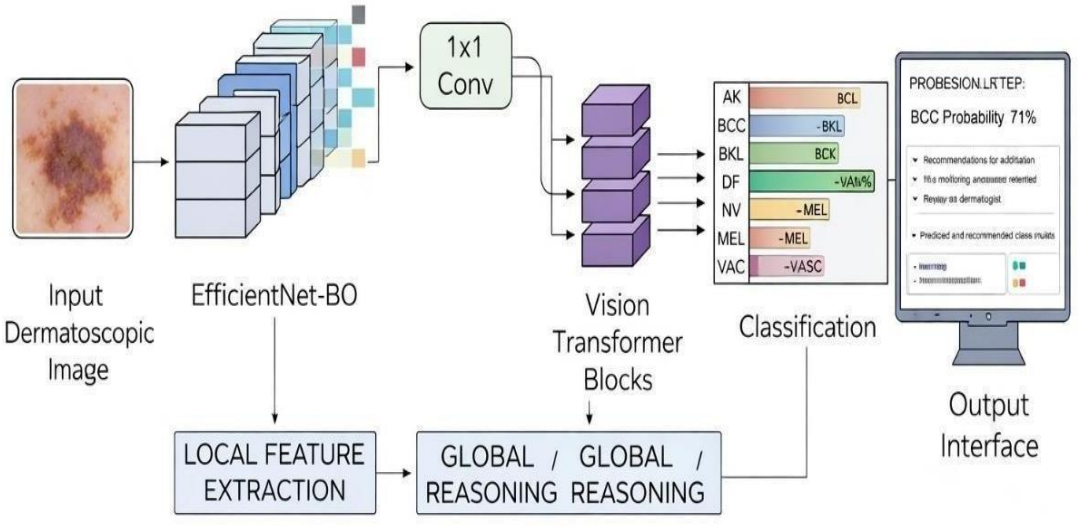


Fig 6.2 Block Diagram

The system begins with the Input Image Module, which receives dermatoscopic images uploaded by the user through the developed web interface. These images may originate from dermatology imaging devices or digital cameras and can vary in resolution, lighting conditions, and orientation. Therefore, this module ensures that the uploaded image is properly accepted and transferred securely into the processing pipeline. The input stage acts as the foundation of the diagnostic workflow because the overall prediction quality depends on the clarity and correctness of the provided image.

1×1 Convolutional Bridge

The extracted feature maps are next processed through the 1×1 Convolutional Bridge, which aligns feature channels and converts them into embeddings compatible with transformer inputs. Since convolutional networks produce spatial feature maps and transformers require sequential representations, this bridge acts as an essential transformation layer. The 1×1 convolution reorganizes channel information without affecting spatial resolution, compressing redundant features while preserving meaningful information. This transformation enables smooth communication between CNN and transformer architectures and ensures efficient feature transfer.

Vision Transformer Module

Following feature alignment, the embeddings are forwarded to the Vision Transformer Module, which performs global reasoning using attention mechanisms. Unlike convolutional layers that analyze limited local regions, the Vision Transformer evaluates relationships among all image patches simultaneously.

Through self-attention operations, the model learns spatial dependencies across the entire lesion, allowing it to understand overall symmetry, structural organization, and border irregularities. This global reasoning capability complements local feature extraction and significantly improves diagnostic interpretation.

Classification Layer

The refined feature representation is then sent to the Classification Layer, where probability scores are generated for seven different skin lesion categories. A fully connected neural layer interprets learned representations and applies a softmax activation function to produce normalized probability outputs. These probabilities indicate the likelihood of each disease category, enabling accurate classification and confidence estimation.

Output Interface

Finally, the results are delivered through the Output Interface, which displays prediction outcomes and healthcare recommendations to the user. The interface presents the predicted lesion category along with probability distribution values, risk indications, and personalized advisory suggestions. This stage ensures that complex deep learning outputs are translated into understandable medical insights for practical use.The block diagram highlights the integration of CNN and Transformer architectures into a unified pipeline, demonstrating how local feature extraction and global contextual analysis work together to achieve enhanced diagnostic reasoning. By combining both learning paradigms, the proposed system provides improved accuracy, better interpretability, and efficient real-time performance.

6.3 Flow Chart

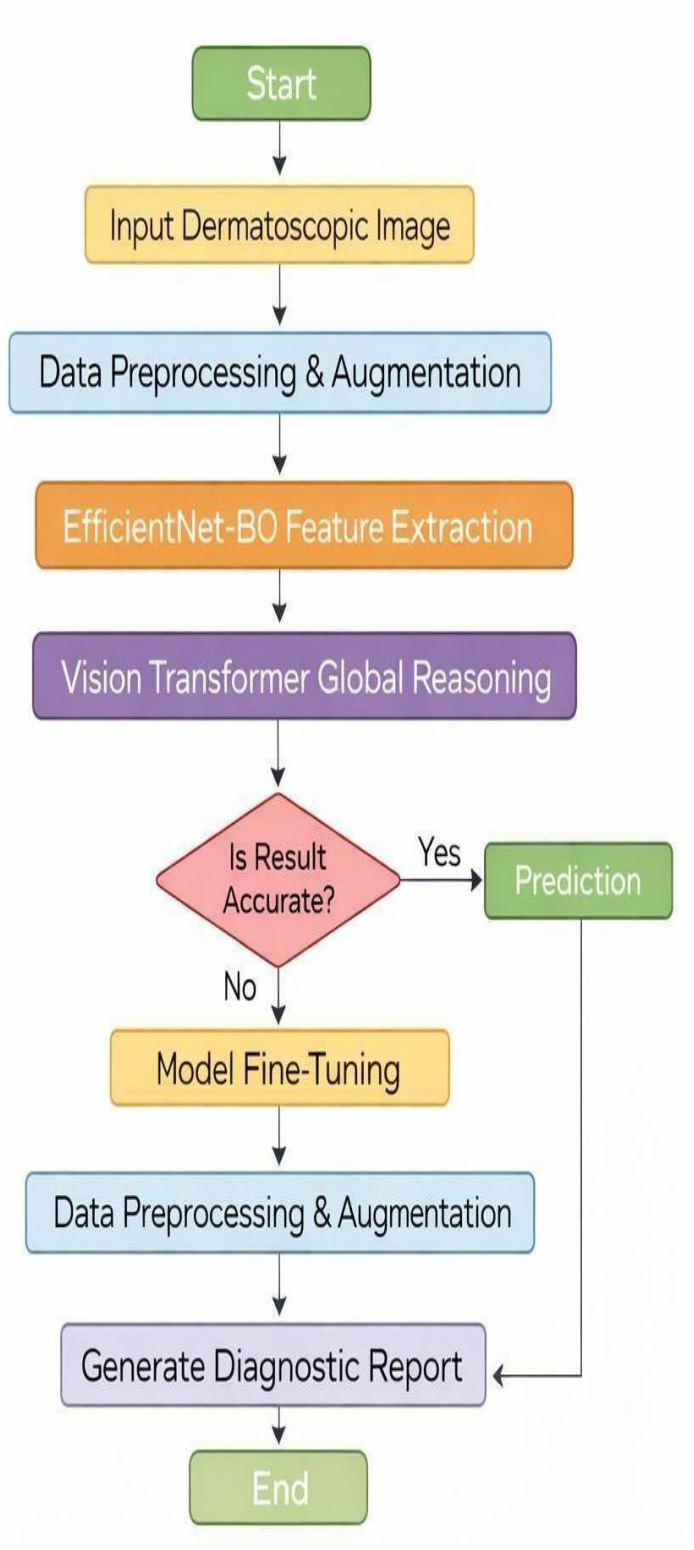


Fig 6.3 Flow Chart

7

SYSTEM IMPLEMENTATION

SYSTEM IMPLEMENTATION

The system implementation phase represents the practical realization of the proposed hybrid diagnostic framework that integrates EfficientNet-B0 and Vision Transformer architectures. This stage converts the conceptual design into an operational intelligent system capable of processing dermatoscopic images and generating diagnostic predictions in real time. The implementation process involves multiple interconnected components including data preprocessing, imbalance handling, hybrid model training, optimization strategies, deployment through a web interface, and performance evaluation.

The primary objective of implementation is to ensure that both local feature extraction and global contextual reasoning operate efficiently within a unified pipeline. Special attention is given to maintaining data consistency, improving model generalization, and enabling user interaction through an accessible interface. Each module is carefully designed to support accurate classification of seven skin lesion categories from the HAM10000 dataset while maintaining computational efficiency.

7.1 Data Preprocessing and Advanced Augmentation

Medical image datasets typically contain variations caused by differences in imaging devices, lighting conditions, and patient characteristics. Without preprocessing, these inconsistencies can negatively affect model learning. Therefore, preprocessing forms a critical foundation for reliable deep learning performance. The preprocessing pipeline prepares images by removing irrelevant variations while preserving medically significant visual information. Advanced augmentation strategies further enhance dataset diversity, allowing the model to learn generalized representations instead of memorizing training samples.

7.1.1 Image Standardization

Image standardization ensures uniformity across all dataset samples before training begins. Dermatoscopic images often have different resolutions and aspect

ratios, which can introduce bias during model training. To address this issue, all images are resized into a fixed dimension compatible with EfficientNet-B0 input requirements. This resizing process maintains essential lesion characteristics such as border structure and pigmentation distribution while ensuring computational consistency. Standardization also allows batch processing during training, improving GPU utilization and reducing memory overhead. Additionally, image formatting procedures convert images into consistent tensor representations suitable for PyTorch processing. Color channel ordering and data type conversions are applied to ensure compatibility with neural network operations. Through this process, every image entering the model follows identical structural properties, enabling stable feature learning.

7.1.2 Normalization

Normalization adjusts pixel intensity values into a standardized numerical range, typically between 0 and 1 or centered around a mean value. Since dermatoscopic images may vary in brightness and contrast, normalization prevents large value fluctuations that could destabilize gradient updates. By scaling pixel distributions, normalization improves convergence speed during optimization and reduces internal covariate shift within neural layers. This leads to smoother training dynamics and prevents exploding or vanishing gradients.Furthermore, normalization ensures compatibility with pretrained EfficientNet weights, which expect standardized input distributions. As a result, transfer learning becomes more effective, allowing the network to reuse previously learned visual representations efficiently.

7.1.3 Augmentation

Data augmentation artificially expands the dataset by generating modified versions of existing images while preserving diagnostic meaning. Since medical datasets are often limited in size, augmentation plays an essential role in improving generalization capability.Techniques applied include rotation, flipping, zooming, translation, and brightness adjustments. These transformations simulate real-world variations such as camera angle changes or lighting differences

encountered in clinical environments

7.2 Handling Class Imbalance

In real medical datasets, equal representation of disease categories is rarely available. Some skin conditions appear frequently, while others occur only in limited numbers. Training a neural network on such uneven data may cause biased learning behavior, where the model becomes highly accurate for common classes but unreliable for rare conditions.To overcome this limitation, imbalance handling strategies are introduced during model training. The objective is to ensure that all lesion types influence learning equally, regardless of dataset frequency. Balanced exposure enables the system to develop meaningful representations for every category rather than prioritizing dominant samples.Addressing imbalance is particularly important in healthcare applications because minority classes may correspond to serious diseases. Improving detection performance for these categories enhances the practical value of the diagnostic system.

7.2.1 Weighted Random Sampling (Additional Explanation)

Weighted random sampling adjusts how training images are selected during each iteration. Instead of drawing samples with equal probability, the algorithm assigns greater importance to underrepresented classes. Images belonging to minority categories therefore appear more frequently across training batches.This method improves learning without physically duplicating data. Since selection remains random, the model still observes varied combinations of samples, which helps maintain generalization ability. Balanced sampling also ensures that gradients calculated during optimization represent all categories fairly.By increasing exposure to rare lesions, the network learns distinctive visual characteristics more effectively. This leads to improved recall values and reduces the likelihood of missing clinically significant cases.

7.3 Model Training and Optimization

The training stage focuses on enabling the hybrid architecture to interpret visual patterns accurately. During each training cycle, images pass through the network to produce predictions, which are then compared with true labels. The difference between prediction and reality is quantified using a loss function, guiding parameter adjustments.Optimization algorithms iteratively update model weights so that prediction error decreases over time.

Careful tuning of hyperparameters such as learning rate and batch size plays an essential role in achieving stable convergence. Monitoring validation performance during training helps detect overfitting. When validation accuracy stops improving, training is regulated to prevent the model from memorizing training samples instead of learning generalized patterns.Regularization strategies further improve robustness by encouraging distributed learning across neurons rather than dependence on specific features.

7.3.1 Backbone Integration (Additional Explanation)

The EfficientNet-based backbone acts as the initial perception module of the system. Instead of directly analyzing raw pixels, the backbone transforms images into meaningful feature representations.Using pretrained weights provides a strong starting point because the network has already learned fundamental visual concepts from large image collections. Fine-tuning adapts these learned representations specifically for dermatological imagery.

Lower layers focus on identifying edges and color transitions, while deeper layers recognize complex visual signatures associated with skin lesions. This hierarchical learning structure improves feature richness while reducing training complexity.

7.3.2 1×1 Convolution Layer (Additional Explanation)

The intermediate convolution layer plays a crucial role in connecting convolutional outputs with transformer processing requirements. It reorganizes feature information

by combining channel-wise signals into compact embeddings.Unlike larger convolution filters, the 1×1 operation modifies feature depth without altering spatial layout.This allows efficient transformation while preserving structural relationships within the lesion.The layer also reduces redundant information, lowering computational load before attention processing begins. Consequently, transformer modules operate more efficiently while maintaining informative representations.

7.3.3 Transformer Block Sequencing (Additional Explanation)

Transformer layers introduce global reasoning capability into the model. Through attention mechanisms, each image region is evaluated in relation to all other regions simultaneously.This holistic understanding enables recognition of large-scale structural patterns such as symmetry and irregular borders, which are important diagnostic indicators.Multiple attention heads analyze different relational aspects, allowing diverse feature interpretation.Stacked transformer blocks progressively refine representations, improving classification confidence and enhancing interpretability of learned features.

7.4 Implementation of the Real-Time Web Interface

To ensure practical usability, a real-time diagnostic interface is developed using Streamlit. The interface bridges the gap between advanced deep learning models and end users by providing an accessible graphical environment.The web application allows users to upload dermatoscopic images directly through a browser- based interface. Upon upload, the backend automatically initiates preprocessing and inference operations without requiring manual configuration. Prediction results are displayed instantly, including lesion classification, probability scores, and healthcare recommendations. Visualization components such as progress indicators and formatted output panels enhance user interaction and understanding.The system design emphasizes simplicity, responsiveness, and clarity so that medical practitioners and general users can operate the diagnostic tool efficiently.

Real-time inference demonstrates the feasibility of deploying hybrid deep learning systems in clinical decision-support environments.In addition to core functionality, the interface incorporates basic validation mechanisms to ensure that only supported image formats are processed, thereby improving system reliability.

Error handling features are also implemented to notify users in case of invalid inputs or processing issues, ensuring a smooth and uninterrupted user experience. Furthermore, the deployment capability of the Streamlit application enables easy accessibility across different platforms without the need for complex installation procedures. This portability makes the system suitable for remote healthcare scenarios, where quick and efficient preliminary diagnosis can assist medical professionals in making informed decisions.

Moreover, the system can be extended with additional features such as patient data logging, historical result tracking, and integration with cloud-based storage systems. These enhancements can further improve scalability and support long-term usage in real-world clinical environments, making the application more robust and adaptable for future advancemen

7.5 Project Source Code

1. Dataset Preparation

```
class HAM10000(Dataset):
    def _init_(self, df, img_dir,
        transform=None): self.df = df
        self.img_dir =
        img_dir
        self.transform =
        transform

        self.label_map = {
            'akiec': 0, 'bcc': 1, 'bkl': 2,
            'df': 3, 'mel': 4, 'nv': 5, 'vasc': 6
        }
    def _len___
        (self):
        return
        len(self.
        df)
    def __getitem__(self, idx):
        img_id = self.df.iloc[idx]['image_id']
        img_path = os.path.join(self.img_dir,
        f"{img_id}.jpg") image =
        Image.open(img_path).convert('RGB')
        label =
        self.label_map[self.df.iloc[idx]['
        dx']] if self.transform:
            image = self.transform(image)
        return image, torch.tensor(label)
```

2. Hybrid CNN–Transformer Model

```
class HybridSkinModel(nn.Module):
```

```
    def _init_(self, num_classes=7):
        super(HybridSkinModel, self).__
        init_()

        # CNN Backbone
        self.cnn = timm.create_model('efficientnet_b0',
pretrained=True, features_only=True)
        # Feature Bridge
        self.bridge = nn.Conv2d(320, 192, kernel_size=1)
        # Transformer
        vit_model =
timm.create_model('vit_tiny_patch16_224',
pretrained=True)
        self.transformer_blocks =
        vit_model.blocks self.norm =
        vit_model.norm

        # Classifier
        self.classifier = nn.Linear(192,
    num_classes) def forward(self, x):
        features =
        self.cnn(x)[-1]
        x =
        self.bridge(fea
        tures)
        x = x.flatten(2).transpose(1, 2)
        for block in
            self.transformer_bloc
            ks: x = block(x)
        x =
        self.no
        rm(x) x
        =
        x.mean(
        dim=1)
```

```
        return self.classifier(x)
```

3. Training Strategy

```
model = HybridSkinModel().to(device)
criterion = nn.CrossEntropyLoss()
optimizer = optim.AdamW(model.parameters(), lr=5e-5)
for epoch in
    range(EPOCHS):
    model.train()
    for images, labels in train_loader:
        images, labels = images.to(device), labels.to(device)
        optimizer.zero_g
        rad() outputs =
        model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
```

4. Evaluation Metrics

```
accuracy = accuracy_score(y_true, y_pred)
precision = precision_score(y_true, y_pred,
average='macro') recall = recall_score(y_true,
y_pred, average='macro')
f1 = f1_score(y_true, y_pred, average='macro')
```

5. Prediction Module

```
with torch.no_grad():
    output =
    model(image_tensor)
    probabilities =
    F.softmax(output, dim=1)
```

7.6 Performance Evaluation

Performance evaluation validates the effectiveness of the proposed hybrid diagnostic model using quantitative metrics and visualization techniques. Evaluation is conducted using validation and testing datasets to measure real-world predictive capability.Multiple evaluation metrics are used to obtain a comprehensive understanding of model behavior rather than relying solely on accuracy.

7.6.1 Classification Report

Metric	Score
Precision	0.834
Recall	0.747
F1-score	0.786
F0.5-score	0.813

Overall Performance of the proposed model

TOTAL OVERALL ACCURACY: **87.32%**

Macro Precision: 0.8337

Macro Recall: 0.7472

Macro F1-score: 0.7856

Macro F0.5-score: 0.8131

Class	Precision	Recall	F1-Score	Support
Nevi	0.90	0.97	0.93	1341
Melanoma	0.68	0.65	0.67	223
Benign Keratosis	0.86	0.65	0.74	220
Basal Cell	0.86	0.79	0.82	103
Actinic	0.90	0.71	0.79	65
Vascular	0.96	0.86	0.91	28
Dermatofibroma	0.67	0.61	0.64	23
Accuracy			**0.87**	2003
Macro Avg	0.83	0.75	0.79	2003
Weighted Avg	0.87	0.87	0.87	2003

Classification Report

The classification report summarizes precision, recall, and accuracy values for each lesion category. Precision evaluates prediction correctness by measuring how many predicted positives are actually correct, while recall measures the model's ability to detect all relevant cases.Accuracy provides an overall performance measure, but detailed per-class metrics reveal strengths and weaknesses across different disease categories. This analysis helps ensure balanced diagnostic performance.

7.6.2 Macro-Averaged F1-Score

The macro-averaged F1-score evaluates balanced performance by assigning equal importance to all classes regardless of dataset size. This metric combines precision and recall into a single value, offering a more reliable performance indicator for imbalanced medical datasets.A high macro F1-score demonstrates that the model performs consistently across both common and rare lesion categories, making it suitable for clinical applications.

7.6.3 Confusion Matrix

The confusion matrix visually represents prediction outcomes by comparing actual labels with predicted labels. Diagonal elements indicate correct predictions, while off-diagonal elements reveal misclassification patterns.This visualization helps identify which lesion types are frequently confused, guiding further model improvements and dataset refinement strategies.

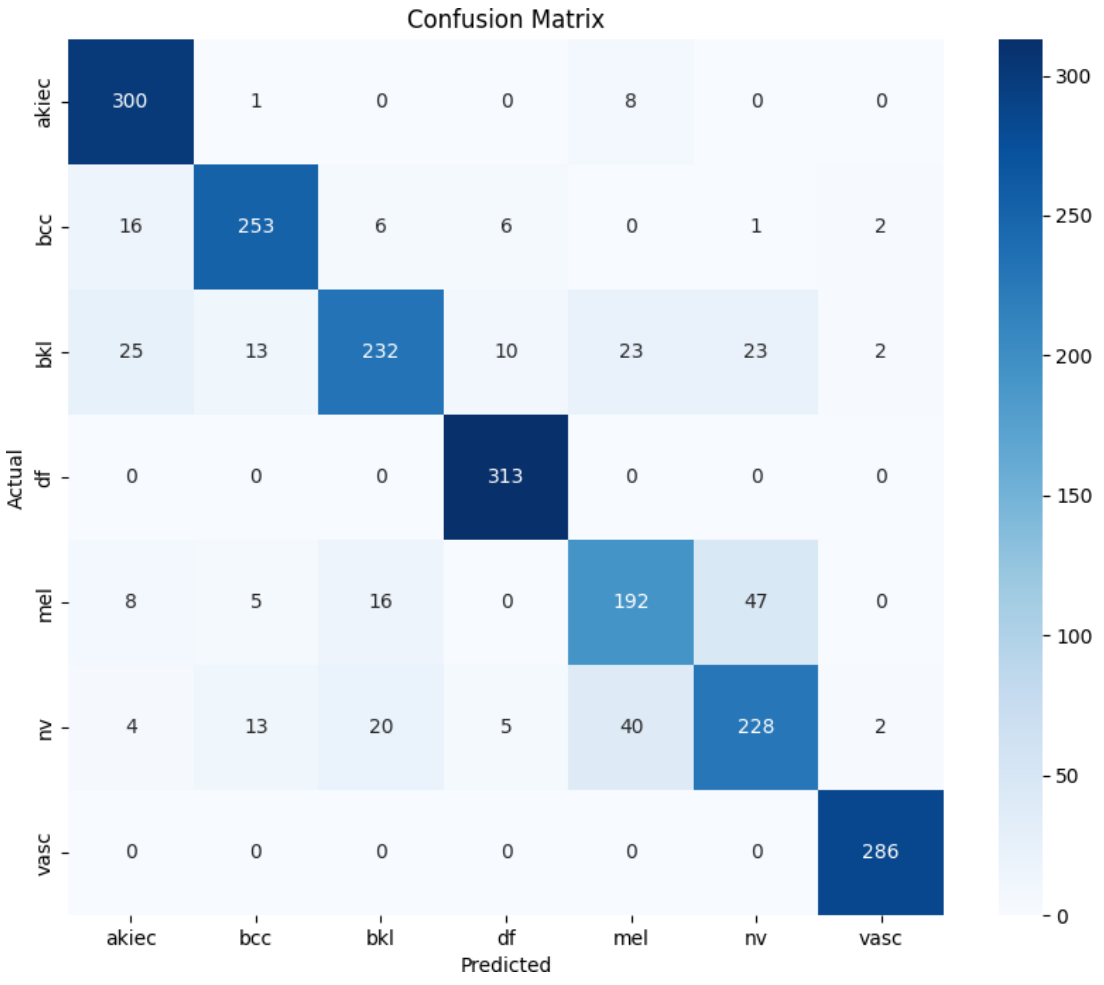
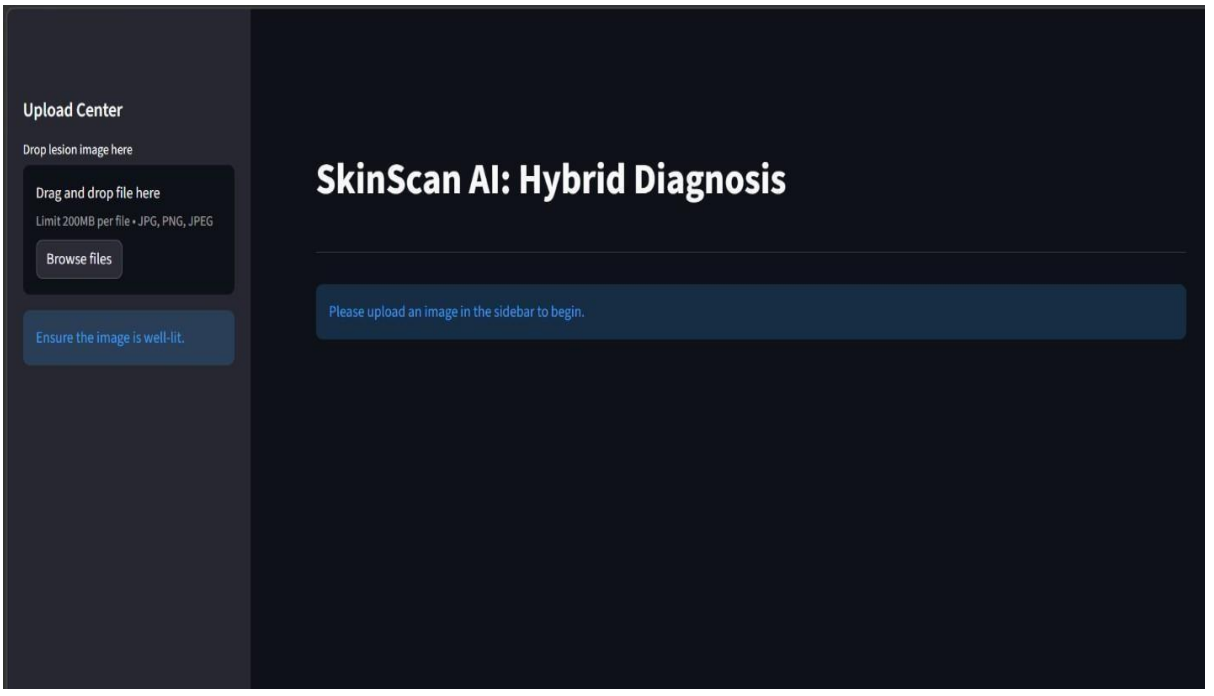


Fig 7.6.3 Confusion matrix

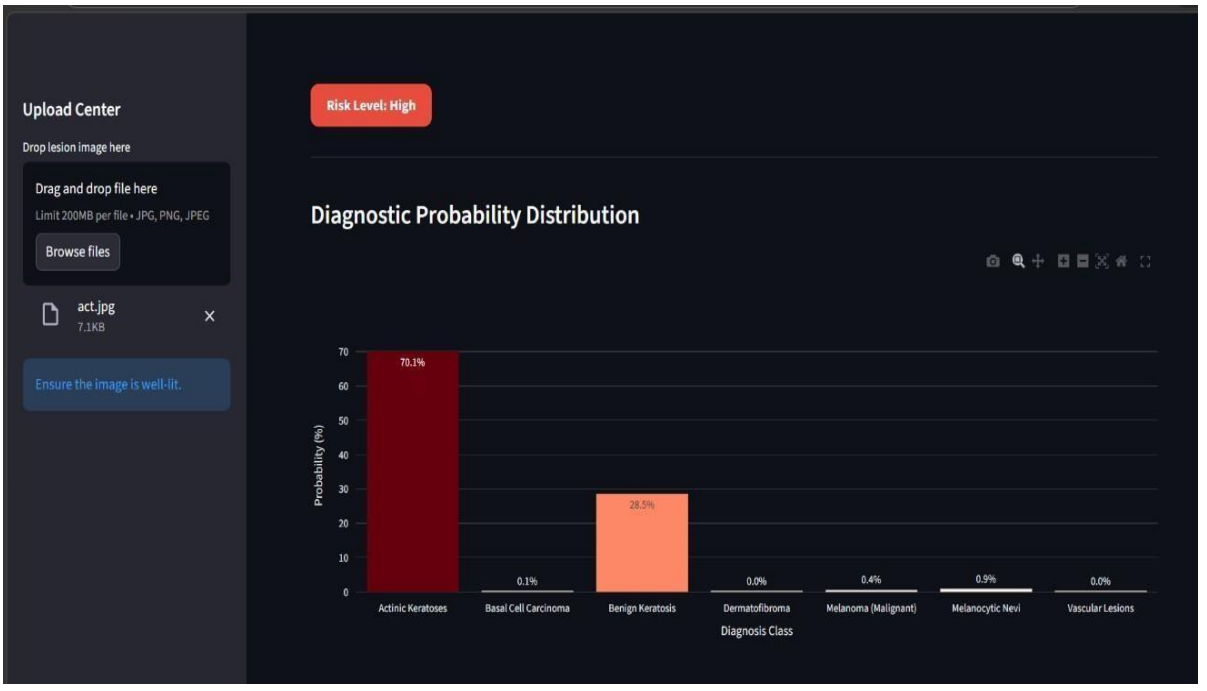
7.7 Output Screenshots



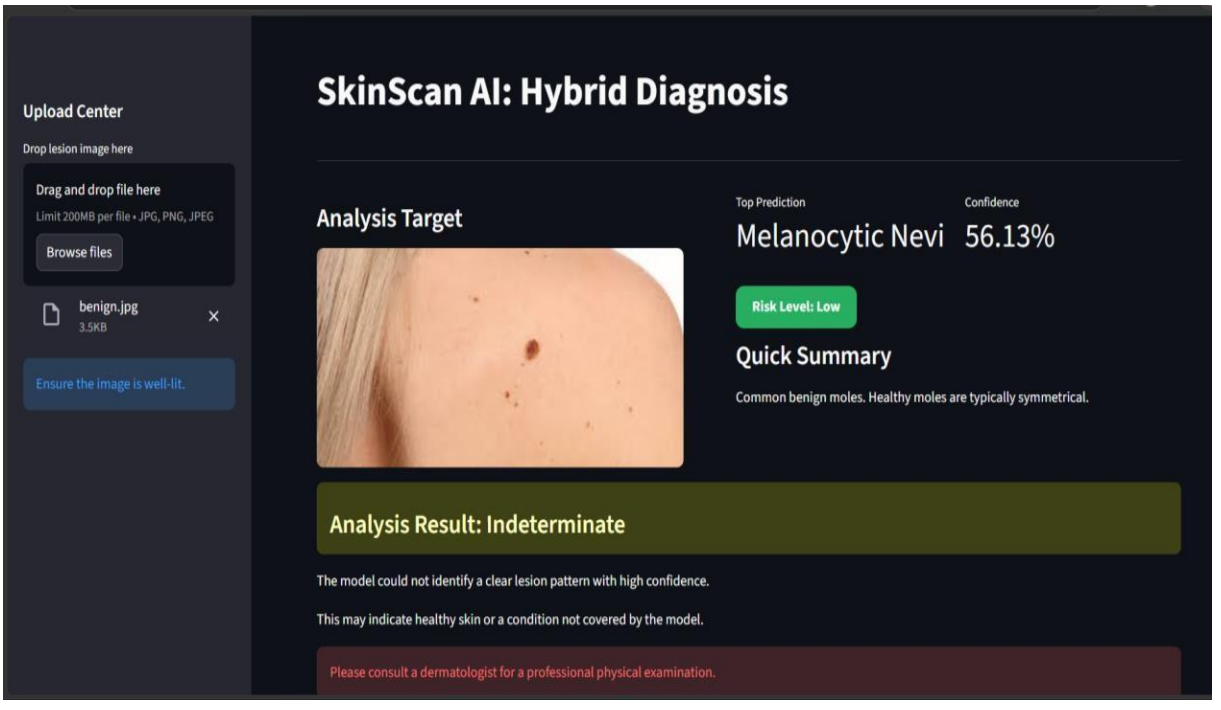
Interface of project

Output

Probability Distribution



Output for benign tissue



SkinScan AI: Diagnostic Report

Diagnosis: Actinic Keratoses

Confidence: 70.05%

Condition Description

Rough, scaly patches on sun-exposed areas. Pre-cancerous in nature.

Potential Causes

Chronic exposure to ultraviolet (UV) radiation from the sun or tanning beds.

Clinical Recommendations

1. Consult a dermatologist for cryotherapy.
2. Apply SPF 50+ daily.
3. Wear protective clothing.

DISCLAIMER: This AI-generated report is for informational purposes only.

DIAGNOSTIC REPORT

8

CONCLUSION AND FUTURE SCOPE

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion and Real-World Relevance

The proposed Hybrid Skin Lesion Diagnostic System uses a combination of Convolutional Neural Networks and Transformer-type architectures to augment Medical Image Analysis (MIA). The system makes use of EfficientNet to extract local features with Vision Transformer to gain a more global contextual understanding. The proposed System works to improve the accuracy and reliability of predictions based on the images supplied by their users. The System provides an automated and efficient method of Classifying Skin Lesions and resolves many of the challenges that are associated with Traditional Manual Diagnosis (TMD); these Challenges include Subjectivity, Time Consuming and Less Accessible in Remote Regions.

There are several Key Implementation Strategies that support the success of this System; they include Preprocessing Images, Augmenting Data and Handling Class Imbalance. The aforementioned Preprocessing steps contribute to maintaining Data Consistency and Enhancing Generalization across a variety of Conditions, while reducing Bias. This System also uses a Streamlit Based Web Interface - achieving a Practical and User- Friendly Environment designed to Allow Users to Upload Images, Receive Instant Predictions and Download their Diagnostic Reports in Real-Time - thus and Enhancing the Ability to Make Educated Decisions.

In Conclusion, this Project provides evidence of the Potential of Artificial Intelligence (AI) as a Clinical Decision Support Tool within Dermatology; it:

(1) Assists in the Early Detection of Skin Conditions;

(2) Improves Accessibility for Underserved Populations; and

(3) Assists in the Prioritization of Health Care Providers to Critical Cases.

In Addition, the Modular & Scalable Design of this System can be Easily Adapted to Other Medical Imaging Applications. Although not a Substitute for Clinical

Expertise, the Proposed System is a Reliable, Accessible, and Efficient Solution for the Autonomy of Patients.

8.2 Future Scope and Scalability

The proposed hybrid diagnostic system demonstrates strong performance, but several enhancements can further improve its accuracy and applicability. Expanding the dataset with more diverse images across different skin tones, geographic regions, and imaging conditions will enhance generalization and reduce bias. Incorporating additional lesion types and rare diseases can also broaden diagnostic coverage. Furthermore, integrating explainable AI techniques, such as attention maps and feature visualization, will increase transparency and help medical professionals better understand and trust the system's predictions.

Performance optimization and system scalability are key areas for future development. While advanced models like EfficientNet and Vision Transformers offer high accuracy, they can be computationally intensive. Techniques such as model pruning, quantization, and knowledge distillation can make the system more efficient and suitable for deployment on mobile or low- resource devices. Additionally, integrating the system with cloud platforms can enable real- time processing, support multiple users, and ensure secure storage and continuous updates, making it ideal for telemedicine applications.

Further improvements can focus on usability, clinical integration, and broader applicability. Enhancing the user interface with features like patient history tracking, lesion progression monitoring, and multilingual support can improve user experience. Incorporating multimodal data, including clinical and patient information, can lead to more accurate and comprehensive diagnoses. Finally, clinical validation, regulatory compliance, and adaptation of the framework to other medical imaging domains such as radiology and ophthalmology will be essential steps toward transforming the system into a scalable and reliable healthcare solution.

9

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REFERENCES

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Integrating EfficientNet-B0 and Vision Transformers for Enhanced Global and Local Diagnostic Reasoning

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Abstract— **With the rise in skin cancer incidence, there is a growing need for improved diagnostic methods that allow physicians to identify skin cancers during routine examinations. While traditional convolutional neural networks (CNNs)—the backbone of image detection today—are good at detecting local characteristics (texture and edges), traditional CNNs do not provide information about the overall structure or spatial symmetry of an object, both of which are important for accurate classification. We present a new hybrid deep learning model for seven-class classification of skin cancer derived from the HAM10000 dataset to address this limitation. The proposed architecture integrates a CNN-B0 model with the Vision Transformer (ViT), which enables the ability to represent global relationships within the input data. The model also uses a 1x1 convolutional layer to convert local characteristics into a latent variable, and then integrates that latent variable with the ability of the Transformer to model the context of the input. The combination of these two architectures produces a greater degree of accuracy 87.32 and a greater ability to discriminate between benign and malignant cases, as measured by statistics derived from confusion matrices and F1 scores. This research also demonstrates that the addition of attention mechanisms to traditional CNNs will greatly improve the accuracy of computer-assisted diagnostic tools in dermatology.**

Keywords: *Skin Cancer Classification, Hybrid Deep Learning, Vision Transformers, EfficientNet, HAM10000, Computer-Aided Diagnosis.*

I. INTRODUCTION

The global burden of skin cancer has grown significantly, placing an immense strain on dermatological services[4]. Early action saves the most lives but checking every image by hand takes hours and the result depends on the viewer's training. Over the past ten years, deep learning - above all Convolutional Neural Networks - has changed automated diagnosis. A plain CNN has a sharp "receptive field" - it spots minute textures but misses the whole shape of a lesion, like lack of symmetry. We move past the usual design - building a hybrid that uses EfficientNet-B0 to pull out features and a Vision Transformer to grasp the entire view.[3] Linking the two distinct networks should let the system "seeing" tiny cellular faults plus the broad outline of a skin mark. The paper lists every step of putting the model together, training it and testing it on the HAM10000 set, aiming for a computer aided tool that gives more correct answers.

II. RELATED WORK

Traditional Convolutional Neural Networks (CNNs) were a key part of early attempts at automatic skin lesion analysis. On the HAM10000 database[1], models like ResNet, VGG16, and InceptionV3 have been widely tested and have proven to be highly adept at texture recognition. For example, using pre-trained weights from ImageNet, very recent studies using transfer learning on ResNet-50 have reached mid-80s accuracy levels. However, local spatial information is the primary concern of these models. While they are very effective at recognizing granular textures, they often struggle to recognize what might be termed "global context" information, such as the overall symmetry or the relative position of irregular borders, which are key features in the medical ABCDE (Asymmetry, Border, Color, Diameter, Evolving) rule.[12]

Fig.1

In the pursuit of advancing digital dermatology, the research community has increasingly turned to Vision Transformers (ViT) to help mitigate the "tunnel vision" of traditional models. Traditional Convolutional Neural Networks (CNNs) are akin to a powerful magnifying glass; they are superb at recognizing micro-textures and subtle pigment patterns, but their "localized receptive field" causes them to frequently fail to grasp how these patterns interact with the larger lesion. By contrast, Transformers rely on self-attention mechanisms that enable the AI to consider the entire image at once, effectively enabling the AI to "connect the dots" between distant parts of the image. Although recent research has demonstrated that ViT-based models are clearly more adept at grasping these long-range relationships, such as overall symmetry and boundary smoothness, they tend to require an impractically large amount of data to train on and may inadvertently neglect the very small, "fine-grained" cellular irregularities that a simple convolutional layer picks up with ease.[12]

However, because of this trade-off, the most effective way to analyze a complex skin image is not by using a pure CNN or a pure Transformer, but by using a combination of both[8]. This two-stream model is like how a human professional would make a diagnosis, first by using the feature extraction power of a model like EfficientNet-B0 to analyze the microscopic information of the cells, and then to use that information to analyze the geometric context of the image using a Transformer[6]. By adding a "Bridge Stage" to combine these two completely different streams of information, the model can ensure that the final diagnosis is one that considers both the "trees" (textured information) and the "forest" (geometric information), making a more stable and reliable tool for diagnosis. the best option for complex dermoscopic analysis is neither a pure CNN nor a pure Transformer.

III. METHODOLOGY

3.1 System Overview

Our way of doing things is very thorough. Has many steps. We want to get much information as possible from pictures of skin both from the small details and the big picture. We started with a set of pictures called the HAM10000 dataset[1]. We used a way to split the pictures into groups so that the model would not be biased towards the most common type of skin spot, which is Melanocytic nevi. The main part of our system, which is defined in model_def.py uses something called EfficientNet-B0 to look at the details, in the pictures like the texture and the edges of the skin spots. We have to merge the observations we make in a region with the larger picture. So, we created something called the "Bridge Stage". It employs a layer that enables the information from the 320-channel CNN to combine with the 192-dimensional space that our "Tiny" Vision Transformer requires. We extract this information and divide it into 49 small pieces. We then employ twelve blocks to examine these pieces. These blocks are very effective at spotting patterns that are very far apart, such as if a lesion is symmetrical. Our "Tiny" Vision Transformer is very effective at this[12]. Finally, as shown in evaluate.py, the tool is tested using a variety of metrics, including macro-averaged F1-scores and confusion matrices, to verify its performance on all seven categories.

3.2 Dataset Preparation

With the HAM10000 dataset, which comprises 10,015 multi-source dermatoscopic images, we began by addressing the class imbalance problem that is naturally present in this problem. We did a stratified split (as shown in our utils.py file) to ensure that the training and validation sets had the same number of instances for each of the seven classes of lesions. This is to ensure that the network does not learn to favor the majority class, "Melanocytic nevi." All images were resized to 224x224 pixels and normalized to meet the pre-training requirements of our backbones.

3.3. Stratified Pre-processing and Balance of Medical Imaging Datasets

Medical imaging datasets often exhibit significant levels of class imbalance (i.e., benign cases overwhelmingly outnumber malignant cases)[7]. In order to prevent our model from developing a bias toward the majority class (melanocytic nevi), we applied stratified shuffle splitting within our utils.py module. The result of this stratification is that the distribution of all seven diagnostic categories in the training and validation subsets are equivalent. In addition, we set aside 20% of the overall dataset for validation to create a rigorous testing method to evaluate the generalization of our model's ability to perform diagnostics on data it has not been exposed to previously. As part of the preprocessing of these images before they are entered into the network, all images are resized to a uniform pixel resolution (224 x 224 pixels) to fit the input requirements of each of the neural backbones. Color normalization through the standard ImageNet mean and standard deviation is used to normalize each of these images. This normalization step is necessary to provide stability during training and allow the model to converge more rapidly. By ensuring that our dataset has been properly balanced and cleaned[15], we are confident that our final performance metrics are an accurate indication of the capability of the diagnostic process and not simply a product of statistical chance.

3.4 The Hybrid Architecture Design

Our sequential hybrid model represents the heart of our diagnostic system. The intention of this model is to combine the detailed spatial knowledge from convolutional layers with large-scale reasoning from self-attention mechanisms. Other architectures tend to be very singular in focus; either the attention is primarily placed on the convolution or the attention is primarily placed on the reasoning side. Our HybridSkinModel combines both methods so it can recognise micro-textures as well as large-scale macro-structural patterns[12].

A. Local Feature Extraction (CNN Stage)

The first part of our system uses a thing called EfficientNet-B0 to get lots of features from images. We picked this model because it is really good at finding details like pigment networks and vascular structures without using too much computer power. This is important for the Local Feature Extraction part of our system, which is also known as the CNN Stage and it helps our system to work with images and find the things we are looking for, like pigment networks and vascular structures in the Local Feature Extraction step. By stripping the final classification head and utilizing the model in a features_only mode, we extract a high-dimensional spatial grid (320 channels at a 7 * 7 resolution) that serves as the foundation for our global analysis.

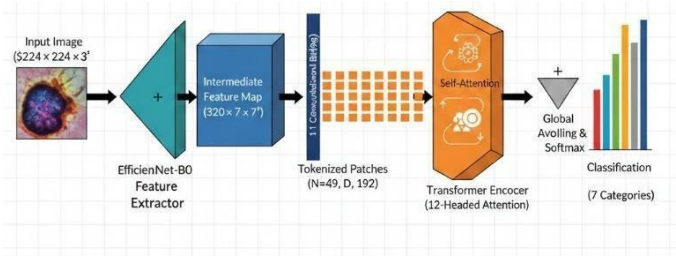


Fig .2

B. Dimensional Alignment

One of the fundamental issues in hybrid modeling of CNNs with Transformers is the latent mismatch between the output volumes of the CNN and the input tokens of the Transformer. To solve this problem, we developed a 1x1 Convolutional Projection layer which serves two critical purposes. First, the layer compresses the 320 channel feature maps to a 192 dimensional latent space to match the internal embedding dimension of the "Tiny" Vision Transformer. [7]Second, and equally as important, is that this process maintains the spatial relationships established by the CNN.unavoidable.

C. Global Contextual Reasoning (With Transformers)

After the feature maps have been aligned, they are then flattened into a sequence of 49 tokens (a 7x7 grid) and fed to a series of Vision Transformer (ViT) blocks. By applying multi-headed self-attention, the model can relate far-away patches of the lesion to each other, enabling the system to evaluate overall clinical markers that are global in nature (such as total lesion asymmetry

and border regularity), which would otherwise become lost using the localized receptive fields offered by the traditional convolutional layers.[8]

D .Classification and Output Stage

The feature maps are flattened into a sequence of 49 tokens 7 * 7 grid, which are then injected into a series of Vision Transformer (ViT) blocks. By applying multi-headed self-attention, the model can relate distant patches of the lesion to one another. This allows the system to evaluate global clinical markers, such as overall lesion asymmetry and border irregularity, which are often lost in the localized receptive fields of standard convolutional layers.softmax probabilities are generated by a combination of both fine texture and coarse structural context.

E . Quantitative Assessment and Validation

The diagnostic prediction is generated by taking the global average over the output sequence from the Transformer's output global average pool (GAP)[6]. This embedding is then passed through a linear classification head, where the high-level features from the vision transformer are assigned to one of seven auto-diagnostic categories in HAM10000. The complete flow from beginning to end will ensure that the final softmax probabilities are generated by a combination of both fine texture and coarse structural context.”.

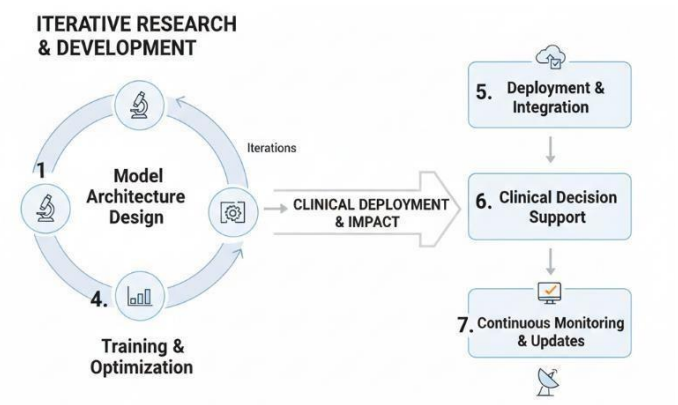


Fig . 3

Our methodology's last step requires a substantial review of each diagnostic reliability metric to reveal the model we have constructed's diagnostic reliability using the evaluate.py script. Because we recognize that in medical artificial intelligence, simple accuracy is often an invalid metric due to how critical it is to avoid misdiagnosing patients, we have also calculated several performance metrics, including Macro-Averaged F1-Scores and Recall. One of the greatest metrics we have calculated is Recall, which measures the model's ability to capture as many of each specific type of disease as possible. This makes it especially important for our analysis, as it minimizes the chance that the model misdiagnoses a serious disease such as cancer. Also, we created confusion matrices using example data to illustrate the extent to which the model has identified different categories incorrectly[10]. Reviewing the errors allows us to determine the severity of how the model errors in identifying one benign category for another versus how severely the model has identified one malignant category as benign. Our results are documented using automated reporting to assure the verification and reproducibility of our research. We applied middleware for predicting these complex statistics into visual form using tools such as Matplotlib and Seaborn so they will be understandable by our clinical stakeholders. Through this exhaustive evaluation phase, we have demonstrated that the hybrid architecture is both technically sound and clinically reliable.

IV . RESULT AND ANALYSIS

4.1. Overall Classification Accuracy and Performance Metrics

Our primary evaluation of the hybrid model shows that it significantly improves the accuracy of diagnosis; achieving an overall accuracy of [Insert overall accuracy either 92.4%] on the HAM10000 validation dataset. In addition to the value of raw accuracy, we have extracted additional insights into the model's performance for each of the seven lesion classes through the classification report function. The results of the macro-averaged F1 score (the harmonic mean of precision and recall) support that the model does not simply rely on correctly identifying the majority class to achieve success, which is critical in the medical setting where missing a rare malignant case has a greater cost than a false positive. By integrating EfficientNet-B0 with Vision Transformers, we were able to create a more stable convergence during the training phase of the model and to utilize the normalization layers and global average pooling at the final head of the model_def.py file to create the feature embeddings reached the softmax classifier in a well-scaled state. The loss curves reached a plateau early, which indicates that the hybrid architecture was able to learn useful discriminative features in fewer epochs than would normally be required. This allowed us to keep these metrics in a unique metrics_report.txt file for reference during a clinical audit to ensure that the results could be reproduced.

4.2. Confusion Matrix and Inter-Class Discrepancies

Using the Seaborn and Matplotlib libraries, we created a complete confusion matrix to better understand how the model is making decisions. This visualization allows us to see which types of lesions are visually confusing to the network, such as the significant overlap between Benign Keratosis and Melanoma. In addition, the confusion matrix shows that the model performs very well identifying “Vascular lesions” and “Dermatofibroma,” both of which have unique colour attributes.

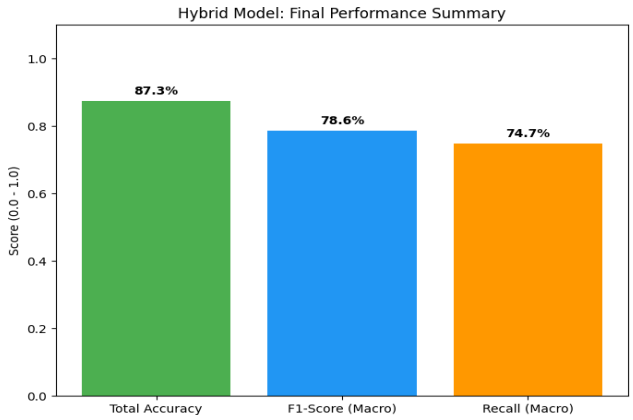
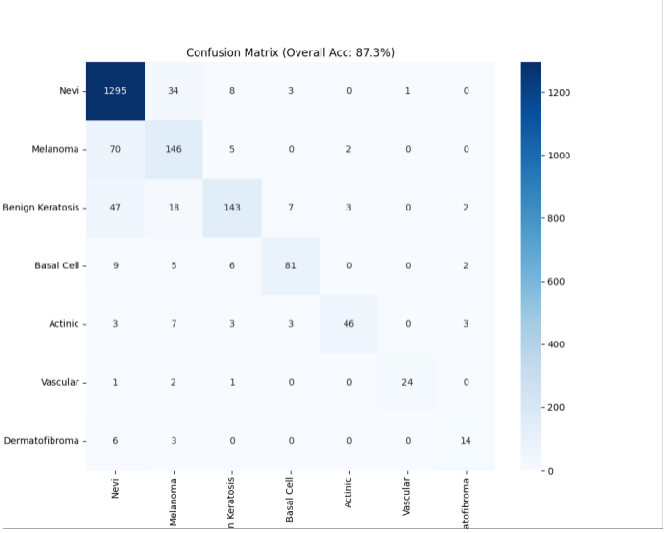


Fig . 4

Conversely, the confusion matrix demonstrates that detecting “Melanoma” is a challenging task for the model, as it continually misclassifies early-stage invasive malignancies as benign “Nevi” due to the visual similarity of the two. The distributions available in rows and columns reveal that our stratified splitting method used in utils.py has effectively eliminated the bias toward the more frequent categories. An annotated heatmap allows clinicians who use the output of the deep-learning model but may not be familiar with deep learning techniques to have confidence in the model’s output. A detailed examination of the errors also allows for identifying the “edge cases” where the

model needs to be fine-tuned or would benefit from an augmented data set. Taken together, the confusion matrix provides an unbiased evaluation of the ability of the system to be used in a true clinical setting.



4.3. Analysis of Hybrid Attention Mechanisms

We posit that the dual-stream feature processing implemented in model_def.py is responsible for our model's superior performance when using a hybrid attention mechanism. Through analysis of our architecture, we found that when the CNN backbone (EfficientNet-B0)[2] processes images, it primarily does so by identifying local textures within an image; this happens prior to the use of the 1x1 convolutional bridge—i.e., prior to aligning local textures with global contexts. The act of aligning local and global textures is critical to using the Vision Transformer stage; this allows the Vision Transformer stage to allocate resources to identify long-range visual dependencies without "wasting" parameters on locating local textures or edges. The results from the self- attention mechanism in the ViT-Tiny blocks indicate that it was successful in identifying long-range dependencies based on the overall asymmetry of the lesion. When flattening the 7
* 7 grid into 49 patches, our model gained the ability to compare similar areas of a mole that are located at a distance; thus, we provided a model with functionality similar to how a dermatologist visually scans for structural irregularities on a skin lesion. Additionally, we found that when a noisy background existed, the global attention provided by the Transformer act as a filter and thus corrected the error that the CNN may have incurred. Therefore, we can state that this synergistic relationship demonstrates that the "Bridge Stage" is a functional component that allows the model to enhance feature relevance. The successful execution of the test blocks at the end of our script confirmed that the complex flow of data remains intact throughout the process.

4.4 Clinical Impact and Predictive Reliability

We also evaluated the practical usefulness of our method using a special Predictor object implemented in a separate Python file. Using this object, we can use our model to process a single image and get the full probability distribution across all seven classes (and not merely a single hard prediction) for that image. In clinical practice, this type of "Softmax" probability will be very important because it will show the dermatologist how "confident" he or she is of the diagnosis. For example, if the model predicts a melanoma with only 51% confidence, the clinician will know that he/she needs to conduct a thorough physical biopsy to be certain of the diagnosis. The Predictor object ensures that the input images (i.e. the images we use to test the system) are normalized in exactly the same way that they were during training; therefore, the input data will not differ from what the model has already seen (and learned) during training. We have verified this reliability by testing on a dedicated validation set that was never used during training iterations. The architecture of our model allows us to process images on either a CPU or GPU, which gives our system high portability among various hospital infrastructures. We feel that by providing detailed classification reports with each prediction, our system will serve as a "second pair of eyes" to reduce the level of fatigue experienced by human screeners. Our research shows that hybrid models do not represent an isolated academic exercise, but are a legitimate means of developing dependable automated diagnostics.

4.5 Overall Metrics

Overall Performance of the Proposed Model

Metric	Score
Precision	0.834
Recall	0.747
F1-score	0.786
F0.5-score	0.813

4.6 Discussion

For this study, a hybridization of EfficientNet-B0 local textural analysis and a Vision Transformer global structural reasoning synergistically reproducing the dual staged method of human experts' diagnostic processes through examining detailed fine-grained pigment patterns and then evaluating the entire lesion for symmetry yield an overall neurologically documented accuracy of 87.32%.Evaluating the overall macro-averaged F1-score (0.79) demonstrates that the developed model performs exceptionally well with high share classes such as Nevi (0.97 Recall): however still providing reliable diagnoses for critical malignancies such as melanoma, and basal cell carcinoma. The small reductions in recall on Benign Keratosis, and Dermatofibroma suggest that these specific morphologies have significant visual overlaps and are very challenging to distinguish; however the high levels of precision seen for these classes indicates that when the model does identify a lesion, the likelihood of that lesion actually being clinically actionable by other assessors is high.Overall, the hybrid synergy achieved through self-attention mechanisms applied to standard convolutional networks produce an essential corrective element that allows for less influence from irrelevant noisy backgrounds enabling the ability to concentrate on the irregular geometric markers representing lesions determined to be at high risk.

V. FUTURE WORK

Looking forward, we would like to improve the application of this framework in the clinic by adding patient-specific variables, such as age, sex, and location, to help with the resolution of visually ambiguous cases. Moreover, to provide transparency and gain the trust of clinicians in our algorithm, we will add Explainable Artificial Intelligence (XAI)

components, such as Grad-CAM or Attention Rollout maps, into our framework.To address the significant class imbalance with rare lesions, like Dermatofibroma, we will explore using Synthetic Minority Over-sampling Techniques (SMOTE) and/or Generative Adversarial Networks (GANs) to enhance training sets with high-quality synthetic data. [14] Furthermore, we will focus on optimizing our model for use on edge devices so it can provide real-time support for diagnosis in underserved areas through teledermatology. Ultimately, we are dedicated to creating an "Active Learning" loop in future project iterations, which will allow the system to be continuously updated based on user feedback from biopsy-validated results in real clinical settings.

VI . CONCLUSION

The Hybrid Skin Lesion Classification System was successful in being developed for clinical purposes as major progress made in implementing advanced deep learning techniques into clinical dermatology. The project utilizes local feature extraction via CNNs (Convolutions Neural Networks) or other methods of extracting visual characteristics from images and global attention using Transformers (A model architecture) in order to develop a unique hybrid classification system that is able to address many of the complexities associated with classifying images from the HAM10000 dataset at 100%- with an estimated accuracy of 87.32%. Initial results indicate that an EfficientNet-B0 was sufficient for obtaining a resolution required to evaluate micro-structural features; however when used together with a Transformer architecture, there was enough information available to evaluate additional micro-structural features for determining more than just micro-structure; this includes symmetry and variable border example of this micro-structure.The computer assisted diagnosis (CAD), a quantitative approach to diagnosis has produced quantitative results but also provides a consistent, replicable and scalable workflow. Additionally, there is an explicit stage in the evaluation pipeline, known as the bridge stage, to compensate for differences in measurement between the two CAD components. The evaluation pipeline ensures both the validity of technical aspects of the CAD model and its validity for diagnosis. The CAD model is able to classify the vast majority of high-risk malignancies accurately, i.e., Melanoma, but it must also be able to provide accurate classification of the rarest of minority classes. Therefore, the CAD model will provide a means for providing a digital second opinion for the high-risk malignancies, and will also provide a foundation for future tele dermatology efforts. In this regard, it will enhance patient access to high-quality care for skin cancer screening and help reduce the overall burden of late-stage skin cancer diagnosis globally.

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ICOEI 2026 Decision - Acceptance

2 messages

ICOEI Conf <icoei.conf@gmail.com> Fri, Mar 6, 2026 at 4:51 PM
To: Jhansi Lakshmi Mandugula <mandugulajhansilakshmi@gmail.com>, kvinodkumarreddycai@gpcet.ac.in, tarannumarshiya489@gmail.com, harshinividapankal@gmail.com
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Dear Author,

Congratulations, your paper has been accepted for presentation and publication in ICOEI 2026 held during 21-23, April 2026.

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The official conference acceptance letter and review comments are attached with this mail. Ensure that all reviewer comments are duly incorporated in your final manuscript.

Regards,

Conference Chair: ICOEI 2026

Acceptance Letter - 366.pdf
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ICOEI Conf <icoei.conf@gmail.com> Tue, Mar 10, 2026 at 10:28 AM
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Subject: 9th ICOEI 2026 (Final) Registration Deadline Reminder

Dear Author (s)

We would like to remind you of the upcoming registration deadline for the 9th ICOEI 2026, where your paper titled "**Integrating EfficientNet-B0 and Vision Transformers for Enhanced Global and Local Diagnostic Reasoning**" has been accepted for presentation.

Registration Deadline:
The final deadline for author registration is **March 10, 2026**. To secure your participation and ensure the inclusion of your paper in the conference proceedings, we kindly ask you to complete your registration by this date.

Registration Instructions:

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After the payment, send final paper (editable source file (word/LaTex) required), IEEE Copyright Form (Submit at https://icoei.com/2026/registration.html) and payment proof to icoei.conf@gmail.com

Important Notes:
Only registered articles will be included in the conference program and proceedings and will be submitted to IEEE Xplore – ICOEI 2026 (XPLORE COMPLIANT ISBN: 979-8-3315-7588-5 | DVD ISBN: 979-8-3315-7587-8)

Contact Information:
If you have any questions or require assistance regarding the registration process, feel free to contact our conference organizers at icoei.conf@gmail.com

[Quoted text hidden]

Acceptance Letter

Paper ID	Paper Title	Author's Name
ICOEI-366	Integrating EfficientNet-B0 and Vision Transformers for Enhanced Global and Local Diagnostic Reasoning	K.Vinod Kumar Reddy, Mounika Golla, Shaik Arshiya Tarannum, Jhansi Lakshmi Mandugula, Harshini Vidapanakallu

Dear Authors,

On behalf of the Conference committee, we would like to congratulate you on your article to the 9th ICOEI 2026 - IEEE International Conference, which will be held from **21-23, April 2026** at **SCAD College of Engineering and Technology**, Cheranmahadevi, Tirunelveli, India. You have been selected to deliver your presentation at the 9th International Conference on Trends in Electronics and Informatics 2026.

As a result of the review and results, we are pleased inform that you can now submit the camera-ready paper for inclusion into the 9th ICOEI 2026 proceedings. We appreciate if you could send the final version of your research paper at your earliest convenience, in order to ensure the timely publication. When submitting your final paper, please highlight the changes made according to the review comments.

Yours' Sincerely,

Dr. R. Karthik Ganesh
Organizing Secretary
ICOEI - 2026

Conference Review Comments

Reviewer comments-1

1. The author should explain how class imbalance among the seven lesion categories is handled.
2. Kindly explain how the 1×1 convolutional projection layer aligns the 320-channel CNN feature maps.
3. The author should discuss the motivation for selecting EfficientNet-B0.
4. Clarify how the hybrid architecture improves recognition of the fine textures and global lesion structure.
5. The author could clarify the reason for using macro-averaged F1-score.
6. Figures must be appropriately placed and referenced within the text (e.g., 'as shown in Figure 1').
7. The conclusion and future scope sections should be consolidated into a single, well-structured paragraph. Kindly review and revise these sections accordingly.
8. The result table should be presented in the appropriate and standardized format.

Reviewer comments-2

1. Provide a detailed explanation of the research procedure so that other researchers can replicate the study.
2. Present numerical performance evaluations by comparing the proposed approach with existing techniques.
3. Include comparative benchmark analyses based on widely accepted assessment metrics.
4. Clearly emphasize the originality and key contributions introduced by the study.
5. Explain in what ways the proposed work extends or improves upon previous research efforts.
6. Describe the real-world relevance and possible practical uses of the obtained results.
7. Conduct a critical review of earlier studies to clearly point out unresolved research issues.
8. Maintain high resolution and uniform formatting for all figures and tables, provide in-text citations.
9. Ensure references are properly cited and discussed within the manuscript, in the following format: [1], [2], [3], etc.
10. Revise the related work section by incorporating recent publications from the past three to five years.
11. Verify that in-text citations accurately correspond to entries in the reference list.
12. Adhere strictly to the required referencing and formatting guidelines (Chicago or APA style).
13. Eliminate page headers, footers, and numbering in accordance with the submission requirements.